

**ATTACHMENT 3 – DESIGN POLICY MEMORANDUM NO. 107 TUNNEL
FINISH DETAILS DATED MARCH 13, 1997**
(40 pages)



InterOffice Memorandum

To: DPM Manual Holders

Date: March 13, 1997

From: A.R. Lancellotti [REDACTED]

File No: EN-1.1.5

Subject: Design Policy Memorandum
No. 107
Tunnel Finish Details

Copies To: J. Bates 01-MX-01

A. PURPOSE

This DPM provides advance information to tunnel finish SDCs regarding in-progress clarifications and changes to CA/T Directive Drawings resulting from the C17AA SDCs design development of the tunnel wall and ceiling details. It also outlines tunnel finish changes as a result of acceptance of certain Cost Containment proposals approved by MHD and FHWA at their Interface meeting with B/PB. The Project intent to date has been that tunnel finish contracts would make use of the detailing by D017A as modified directives for other tunnel sections. Given the lead time before the completion of C17AA, it is important to transmit certain information now to avoid unnecessary, redundant or late work by other SDCs and to allow time to assess how the C17AA details may have to be modified to fit other sections' site specific requirements. It is not expected that CA/T Directive Drawings will be revised; the final detailing from C17AA will be transmitted at a later date by another DPM.

B. Implementation Guidelines

SDCs receiving this DPM shall review the attached materials and plan on implementing similar details to the extent possible in their final tunnel finish designs. Modifications may be required for site specific conditions. Detail changes, if any, to underlying mainline construction contracts should be identified as soon as possible to allow for modifications to those design or construction contracts.

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March 13, 1997
Page 2

C. Contracts Affected

The above guidelines shall be implemented immediately by the SDCs for D009B, D011A, D015A, and D018A.

ARL/kco

Recorded



3-17-97

y. Kalla Date
Change Order Control

Attachments: Summary Sheet plus Sketches numbered 1 to 35.

SUMMARY SHEET FOR THE ATTACHMENTS TO DPM No. 107

C17AA/AB Design Development: Clarifications to CA/T Directive Drawings &
Implementation of Cost Containment Proposals
Related to Tunnel Finishes

Tunnel Finish Issues and Information Available Now:

1. Cost Containment Changes:

Changes & Clarifications to attachment details for
ceramic tile faced precast concrete wall panels.

Sketches: See [1] to [6] for C17AA/AB
detail development to date

2. Cost Containment Changes:

Deletion of tunnel ceiling edge panels - those beyond
the traveled way, except where needed for wider plenum
at or near vent buildings.

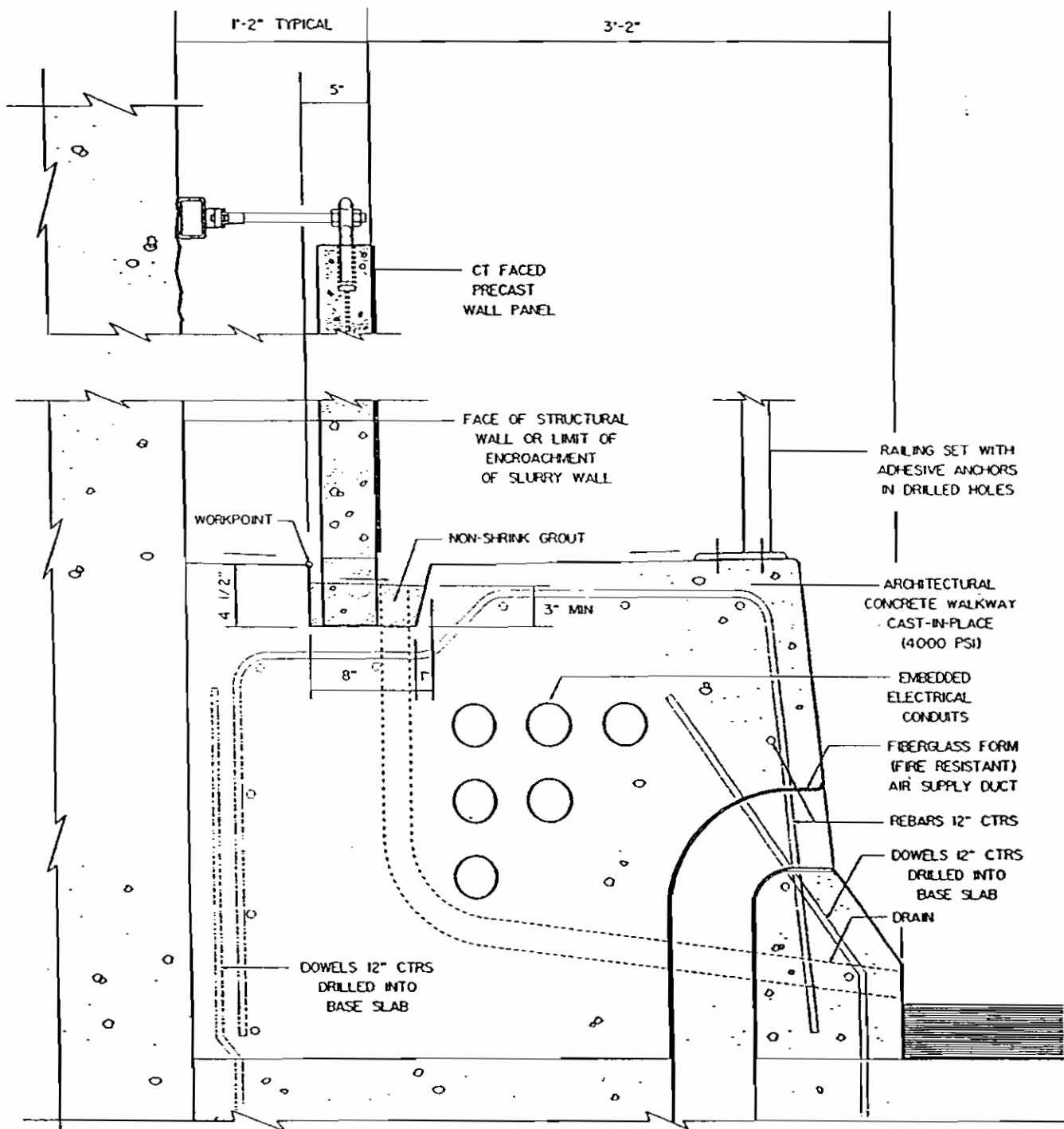
Sketches: See [7] for illustration of deletion.

3. Clarification/Changes to CA/T Directive Drawings:

Changes to the ceiling support system and fireproofing
extent/details developed in response to constructibility
& schedule issues and lessons learned from the TWT.

Sketches: See [8] to [35] for C17AA/AB
detail development to date

FYI - C17AA/AB next submittal is P S & Q and is now scheduled for 05/16/97.

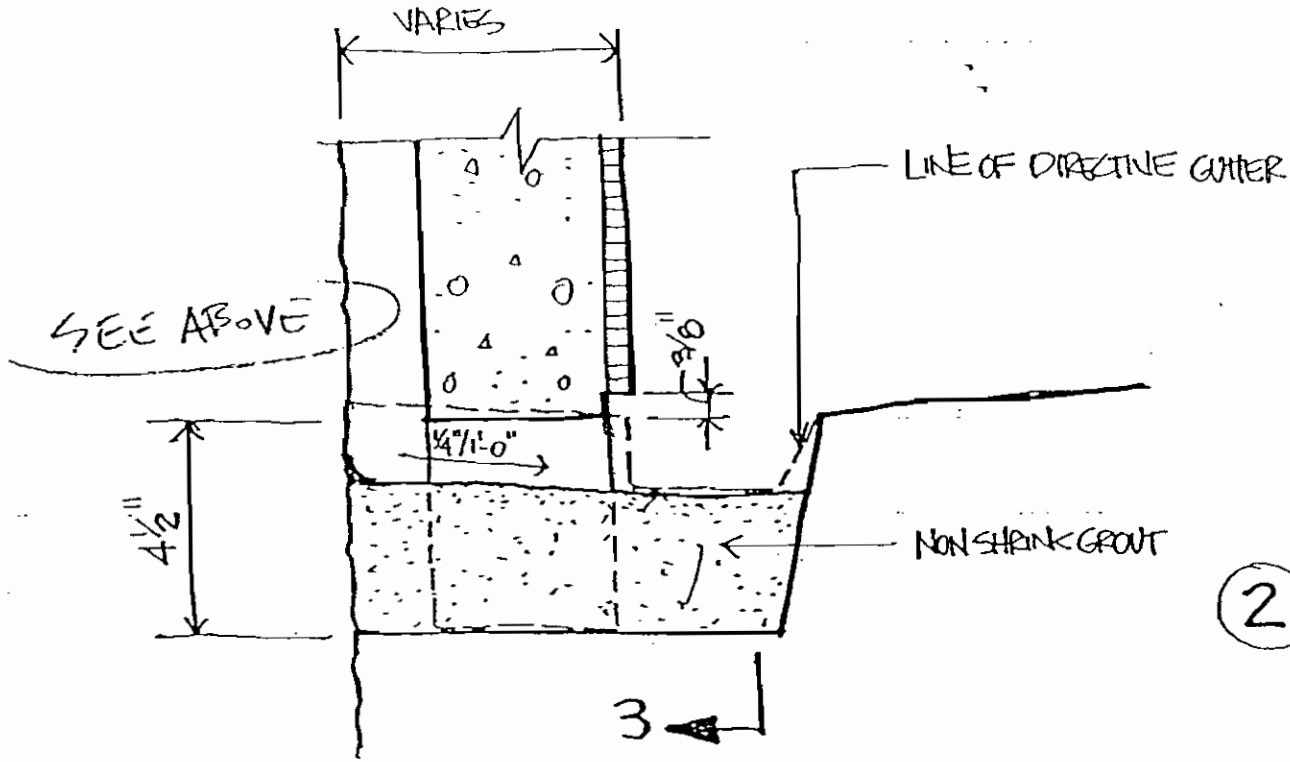
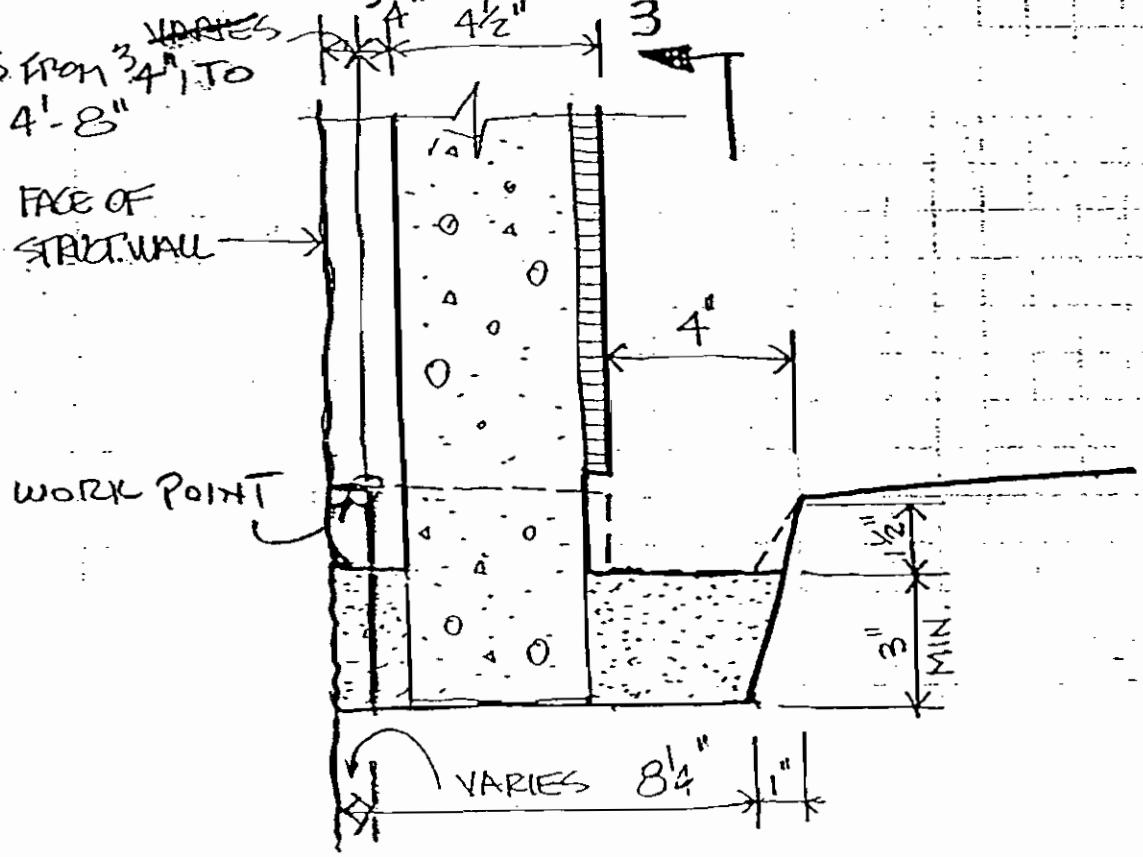


1. TUNNEL WALK/VEHICLE BARRIER

CURRENT PROPOSED ACTION

Made by	Date	Job Number
Checked by	Date	Sheet Number
Backchecked by	Date 11-14-96	2

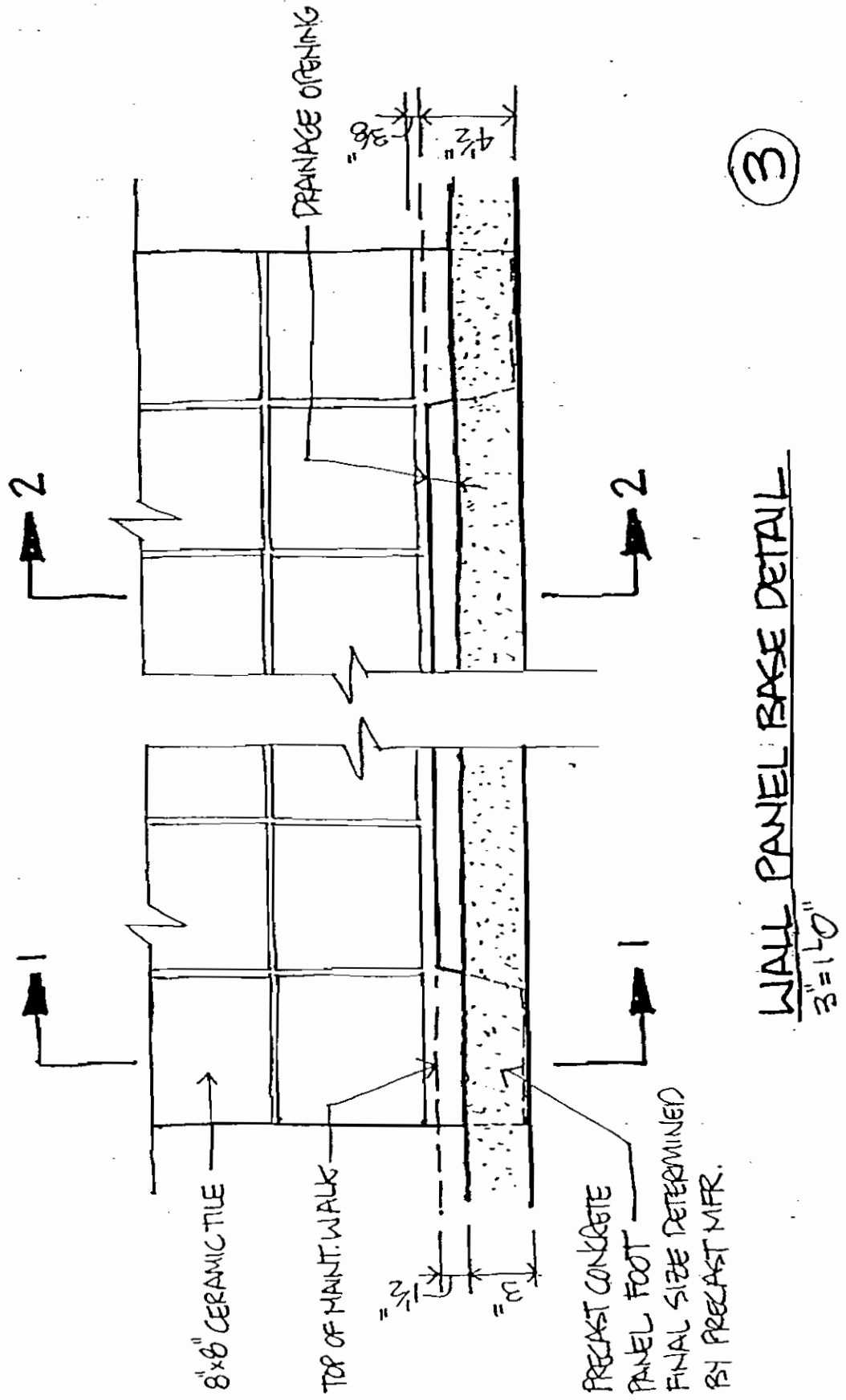
C17AA/AB DETAILS



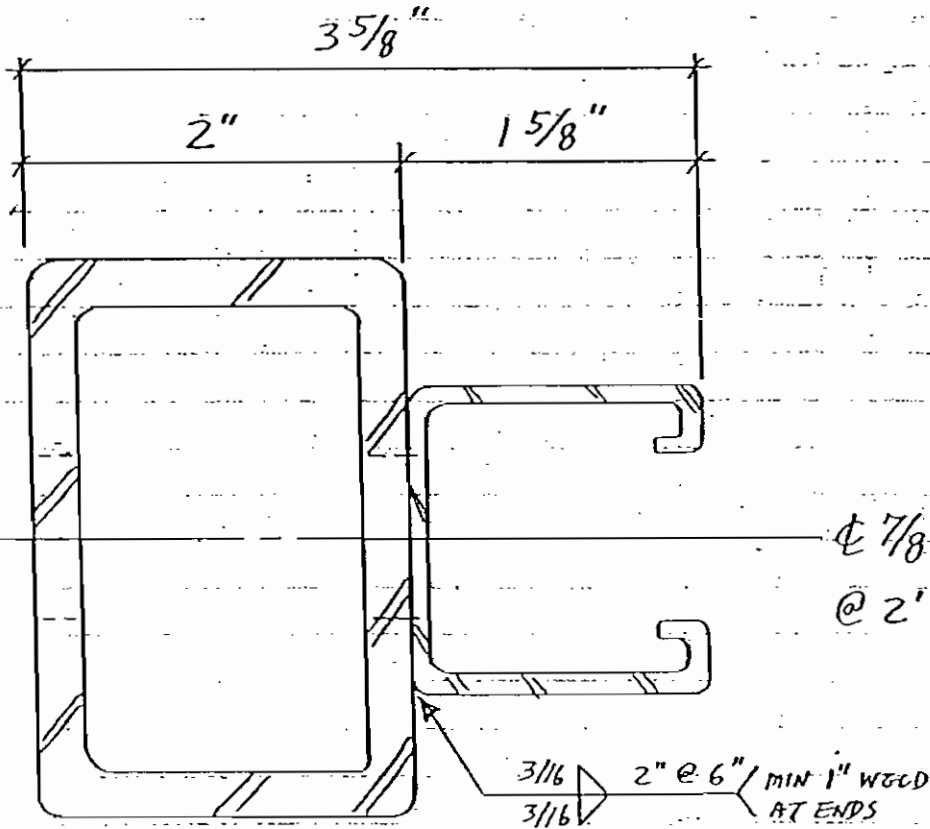
WALL PANEL BASE DETAILS

3'-1'-0"

HNTB The HNTB Companies For <u>CTPA/AB DETAILS</u>	Made by _____	Date _____	Job Number _____
	Checked by _____	Date _____	Sheet Number <u>3</u>
	Backchecked by _____	Date <u>11-14-90</u>	



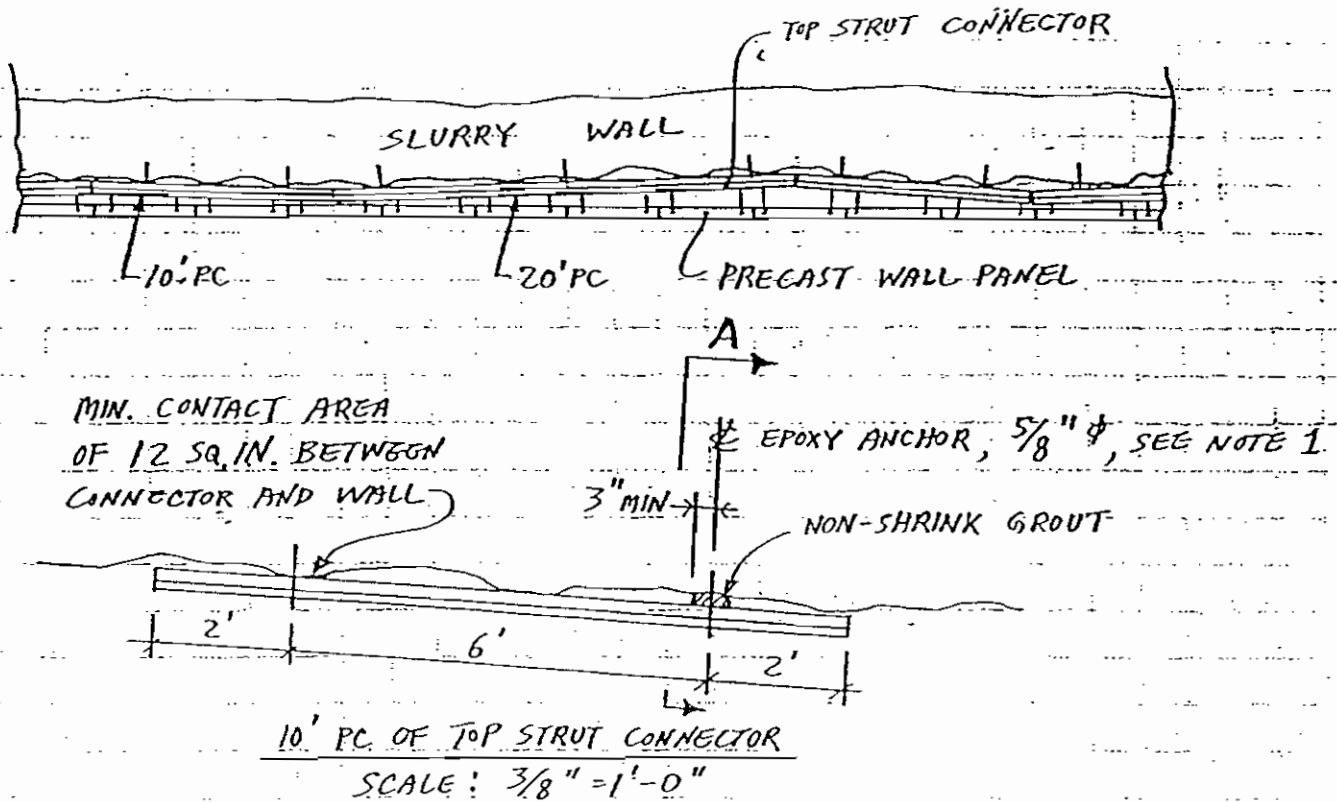
	Made By: [REDACTED]	Date: 1/29/97	Job Number
	Checked By:	Date:	QM-081
For WALL PANEL SUPPORTS	Backchecked By:	Date:	Sheet Number 4



P1000 UNISTRUT ON TS 3X2 X 0.25

TOP STRUT CONNECTOR

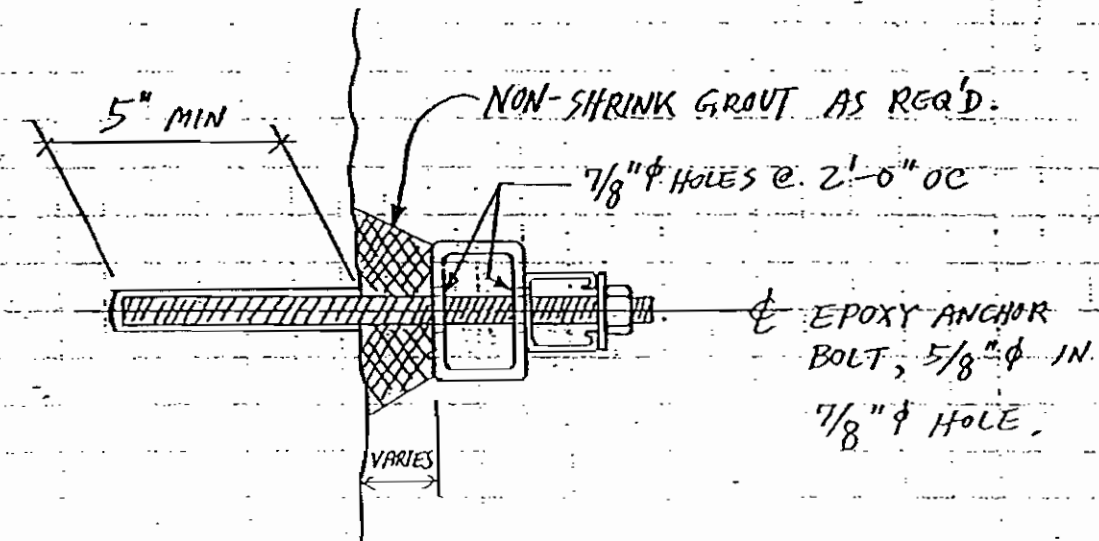
FULL SIZE



NOTE 1 STANDARD UNIT OF TOP STRUT CONNECTOR SHALL BE 10' OR 20', 2 ANCHORS MINIMUM FOR 10' UNIT AND 3 ANCHORS MINIMUM FOR 20' UNIT. MAXIMUM ANCHOR SPACING SHALL BE 8'-0" O.C. MAXIMUM LENGTH OF END CANTILEVER SHALL BE 2'-0".

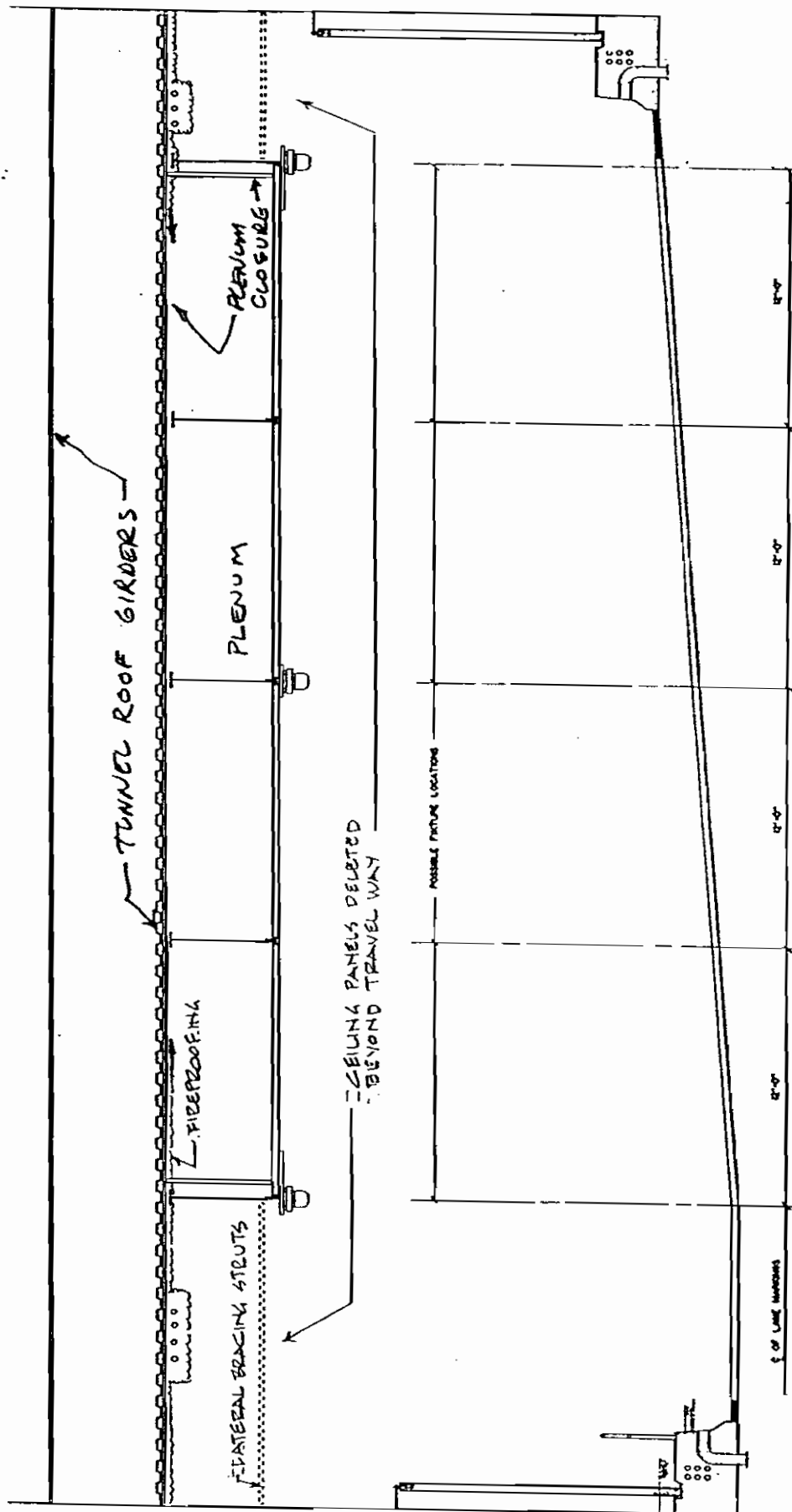
NOTE 2 NON-STANDARD LENGTHS OF TOP STRUT CONNECTORS SHALL HAVE A MINIMUM OF 2 ANCHORS PER UNIT, MAXIMUM ANCHOR SPACING OF 8'-0" O.C., AND MAXIMUM END CANTILEVER LENGTH OF 2'-0".

	Made By: [REDACTED]	Date: 1/30/97	Job Number
	Checked By:	Date:	QM-081
For WAIL PANGU SUPPORTS	Backchecked By:	Date:	Sheet Number 16



SECTION A-A

SCALE: 3" = 1'-0"



TUNNEL FINISHES
 CONCEPT DIAGRAM
 T/MAR 07 B/BIWFA

7

CEILING PANEL SUPPORT SYSTEM

Tunnel Finishes - Revisions To Tunnel Ceiling Support System

Relative to constructability issues of the precast concrete ceiling system, the proposed system herein will allow direct installation of the 40-foot modular ceiling units without tilting them. This new system is created for ease of installation as well as requiring the least amount of installation time per unit.

Each 40-foot by 12-foot modular unit is comprised of ten (10) 4' by 12' by 3 1/2" thick precast prestressed concrete panels framed by a L 8x6x3/4 support angle along each of the two 40-foot edges. These support angles are attached directly to a 40-foot long hanger support plate suspended on 4 vertical hangers of threaded rods, turnbuckles and clevises.

Within the cross section of the tunnel, the 12-foot modular units will match the 12-foot lanes. The two edges between the walls and the lanes will be open - no ceiling panels. Within these two areas, the ceiling panels are replaced with square structural steel tubes. The spacing of these tubular struts is 8'-0" on centers, matching the joints of the 4-foot panels. The struts are used to transmit the design lateral seismic forces into the tunnel walls as well as providing an architectural feature and possibly support for edge lighting in the tunnel.

The sketches of details included herein are results of the preliminary design effort. The following are design parameters and assumptions used:

- Design live load is a uniform load of 25 psf. Live load impact is not considered.
- Dead load of panels is based on normal weight concrete.
- Seismic load is in accordance with the criteria stipulated under the supplemental specifications of CA/T.

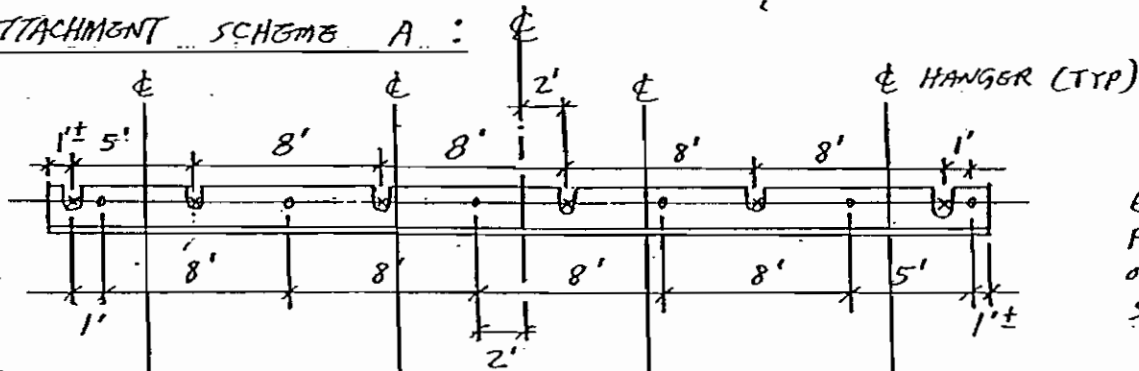
- Design suction pressure of 42 psf is applied at the inside of the exhaust plenum.
- Longitudinal seismic forces will be transmitted by diagonal restraints between the ceiling panel support system and the tunnel roof girders.
- Transverse seismic forces will be transmitted by the ceiling panel support system through the tubular struts into the side walls (slurry or CIP walls).

Installation Procedure For The Modular Ceiling Support System:

1. Install and properly align the hanger support plates and the associated hanger hardware from the tunnel roof girders.
2. The ten (10) panels are preassembled onto the two (2) panel support angles fastened together with the lifting angles by bolts at 24" oc.
3. Transport the modular units to the job site. Each modular unit is to be lifted by the eight (8) lifting lugs provided. All eight (8) points must be lifted simultaneously.
4. At the installation site, each unit is transferred onto a special vertical lift and lifted into position between a pair of hanger support plates. If not uniformly supported on the vertical lift, a minimum of eight (8) support points matching the lifting lugs are required.
5. Once into position, the single shear bolts are placed and tightened, connecting the panel support angles to the two (2) associated hanger support plates. The support system is flexible enough so that the units can be shifted slightly to accommodate construction tolerances.
6. After all lanes of ceiling panels have been installed, the tubular struts at the two (2) edges by the walls are installed. These struts are designed to take up any slack in the ceiling system in the transverse direction due to build in construction tolerances.

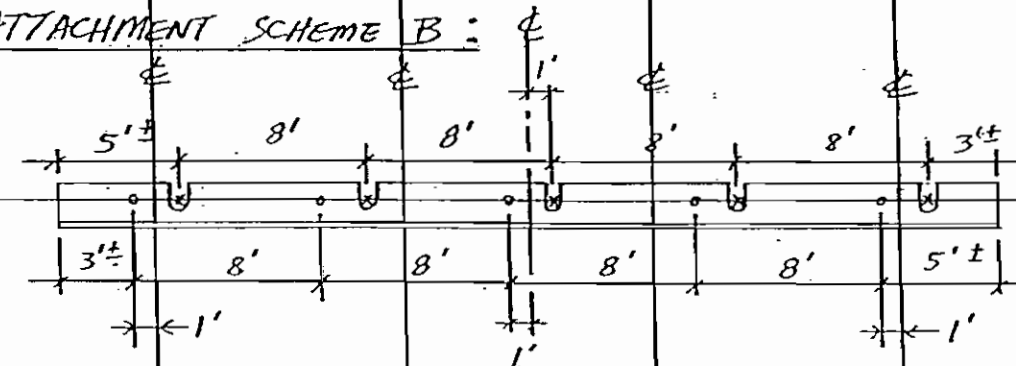
FOR A 40' MODULAR SUPPORT UNIT -

ATTACHMENT SCHEME A :



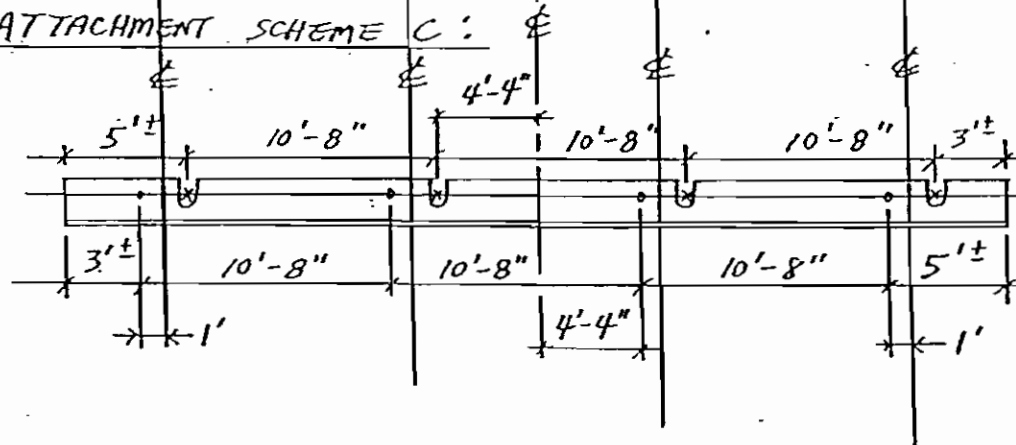
6 BOLTS
PER SIDE
OF ANGLE
SUPPORT

ATTACHMENT SCHEME B :



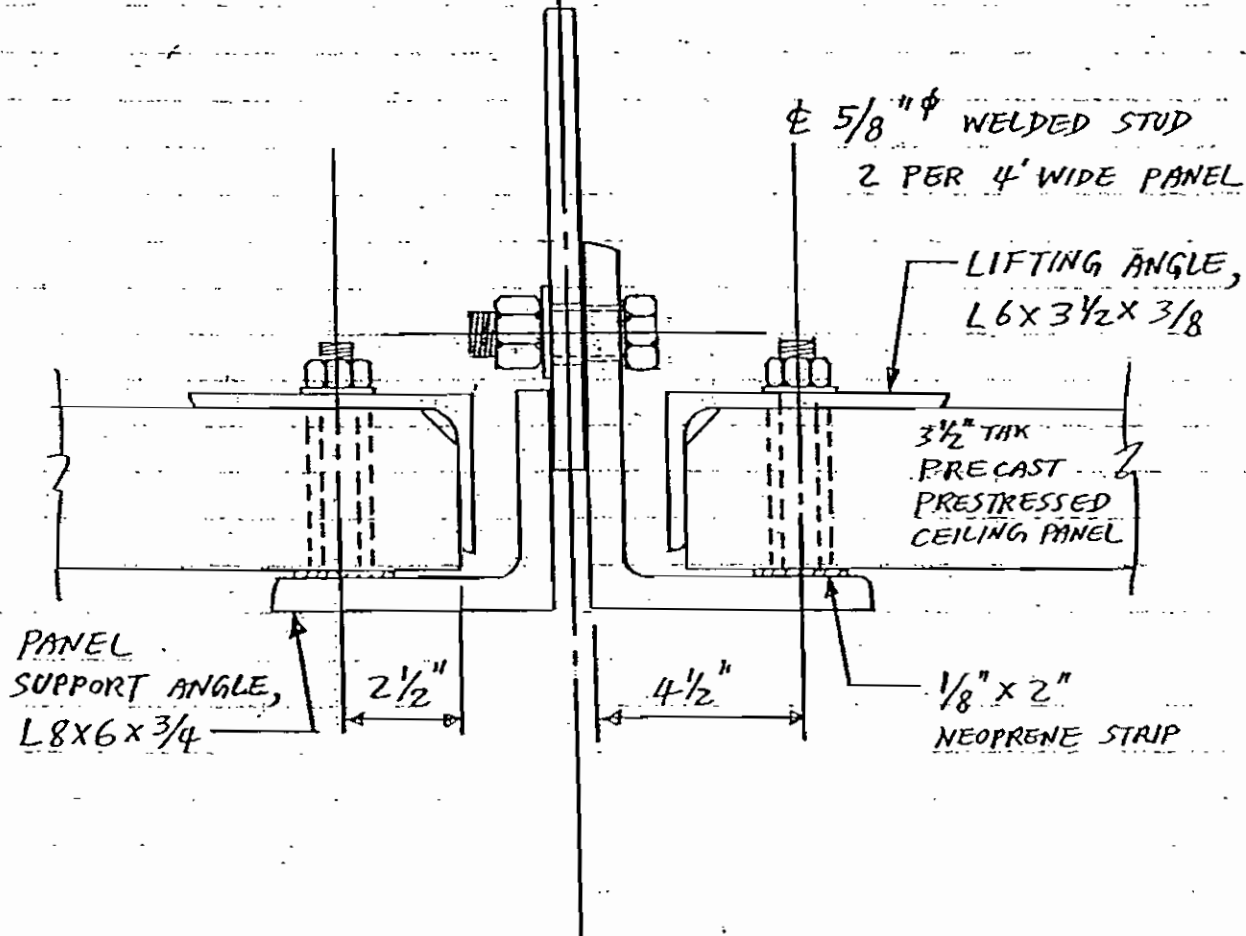
5 BOLTS
PER SIDE
OF ANGLE
SUPPORT

ATTACHMENT SCHEME C :



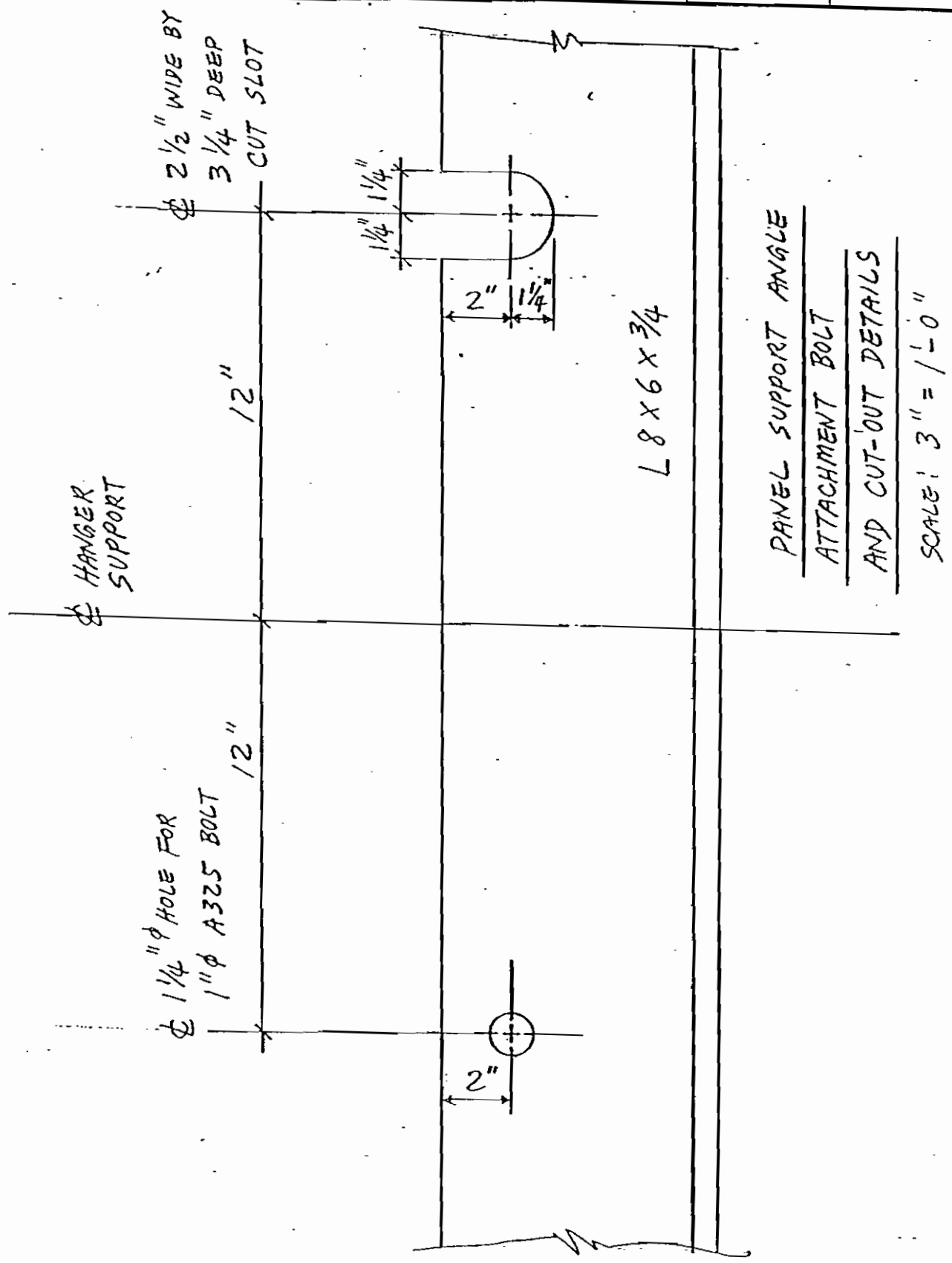
4 BOLTS
PER SIDE
OF ANGLE
SUPPORT

⌀ HANGER SUPPORT PLATE



TYPICAL SECTION
SCALE: 3" = 1'-0"

ES&NB A JOINT VENTURE For CEILING PANEL SUPPORTS	Made By: [REDACTED]	Date: 2/3/97	Job Number
	Checked By:	Date:	QM-081
	Backchecked By:	Date:	Sheet Number 12



Made By: [REDACTED]

Date: 2/3/97

Job Number

Checked By:

Date:

QM-081

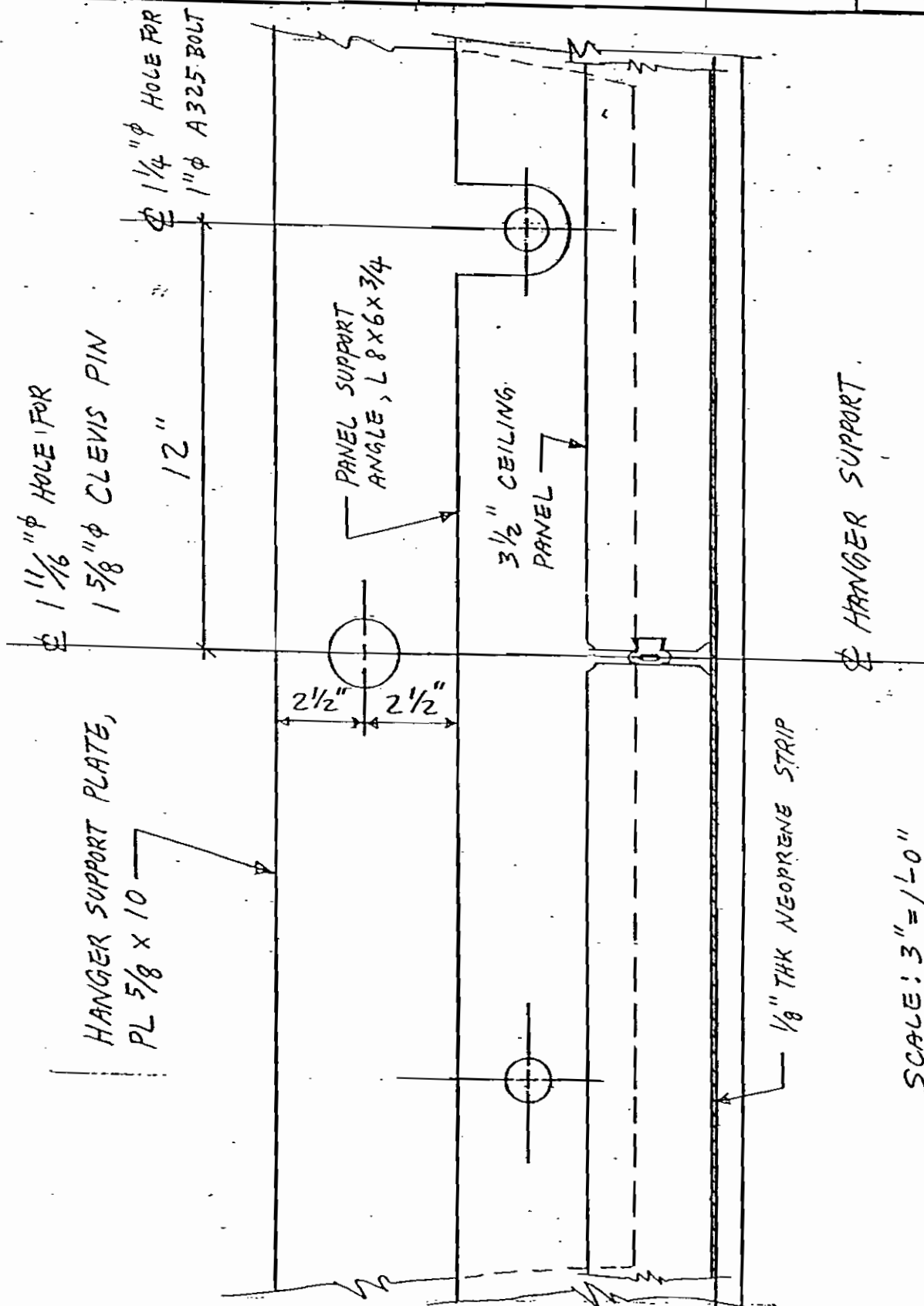
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Date:

Sheet Number

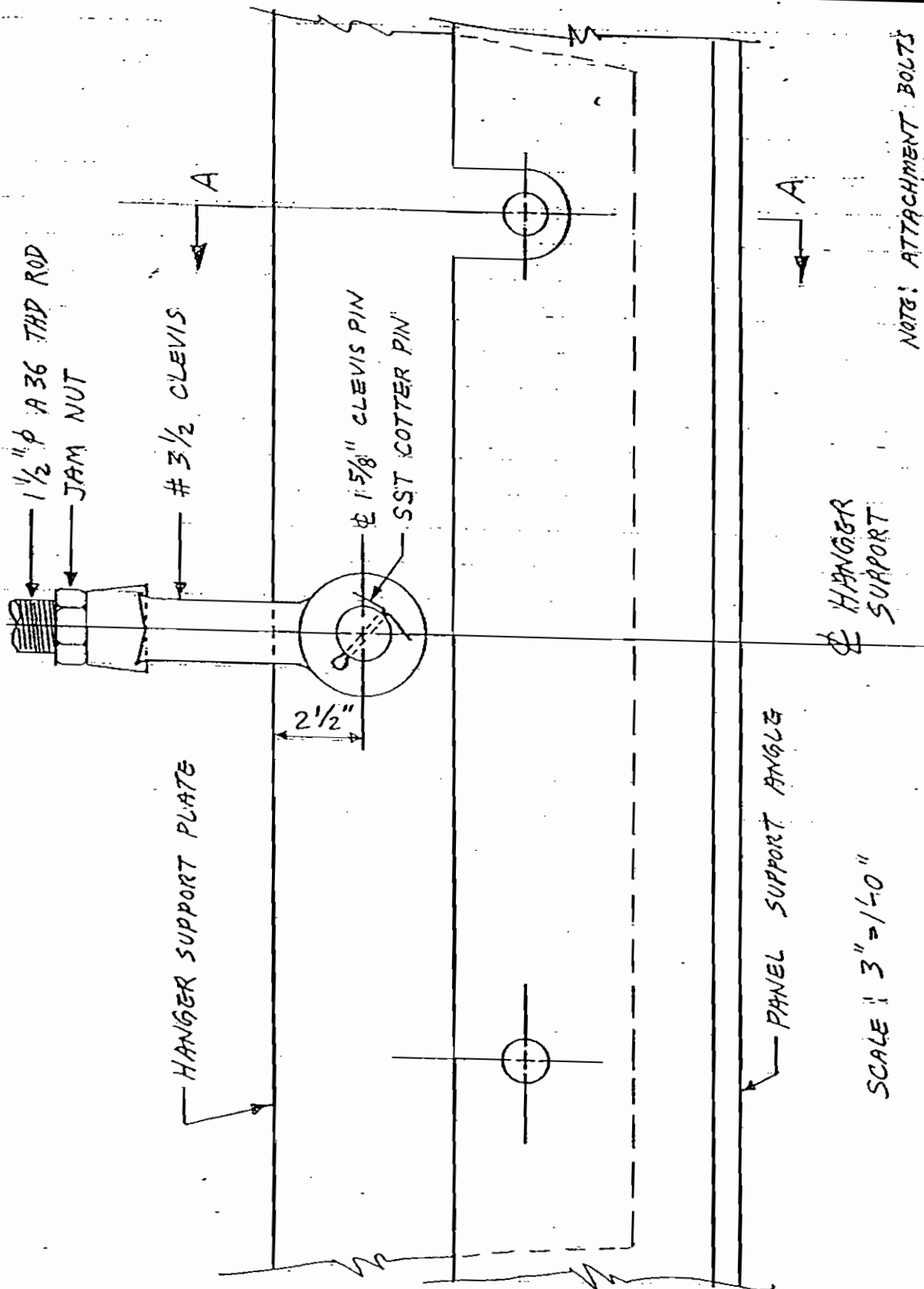
13

For CEILING PANEL SUPPORTS



SCALE: 3" = 1'-0"

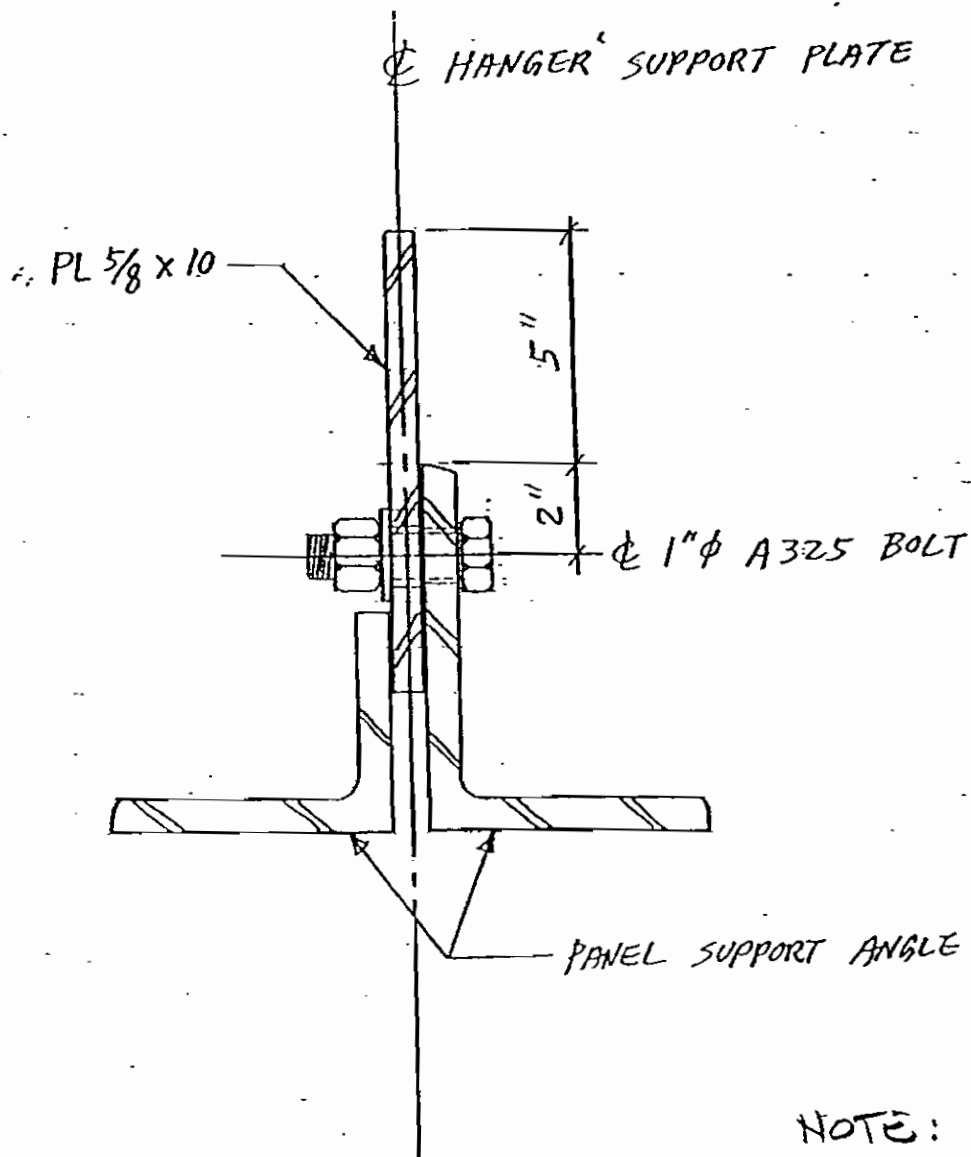
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	Backchecked By:	Date:	Sheet Number 14



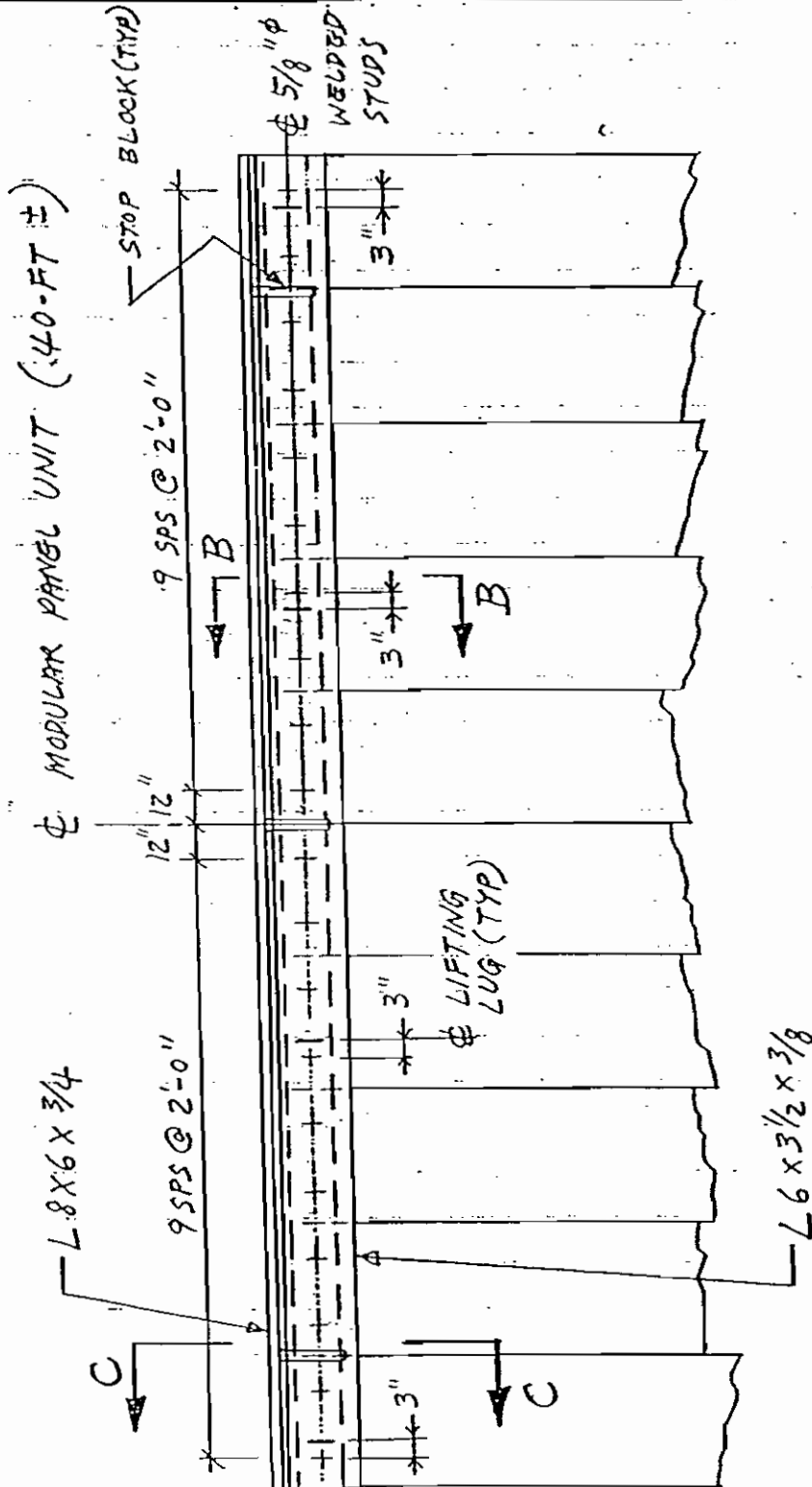
NOTE: ATTACHMENT BOLTS
AND 3 1/2" CEILING
PANELS NOT SHOWN
FOR CLARITY.

SCALE: 3" = 1'-0"

ESTAR A JOINT VENTURE	Made By: [REDACTED]	Date: 2/3/97	Job Number
	Checked By:	Date:	QM-081
	Backchecked By:	Date:	Sheet Number 15



NOTE:
 STRUCTURAL T IS
 BEING CONSIDERED
 FOR OUTER EDGE
 CONDITION.



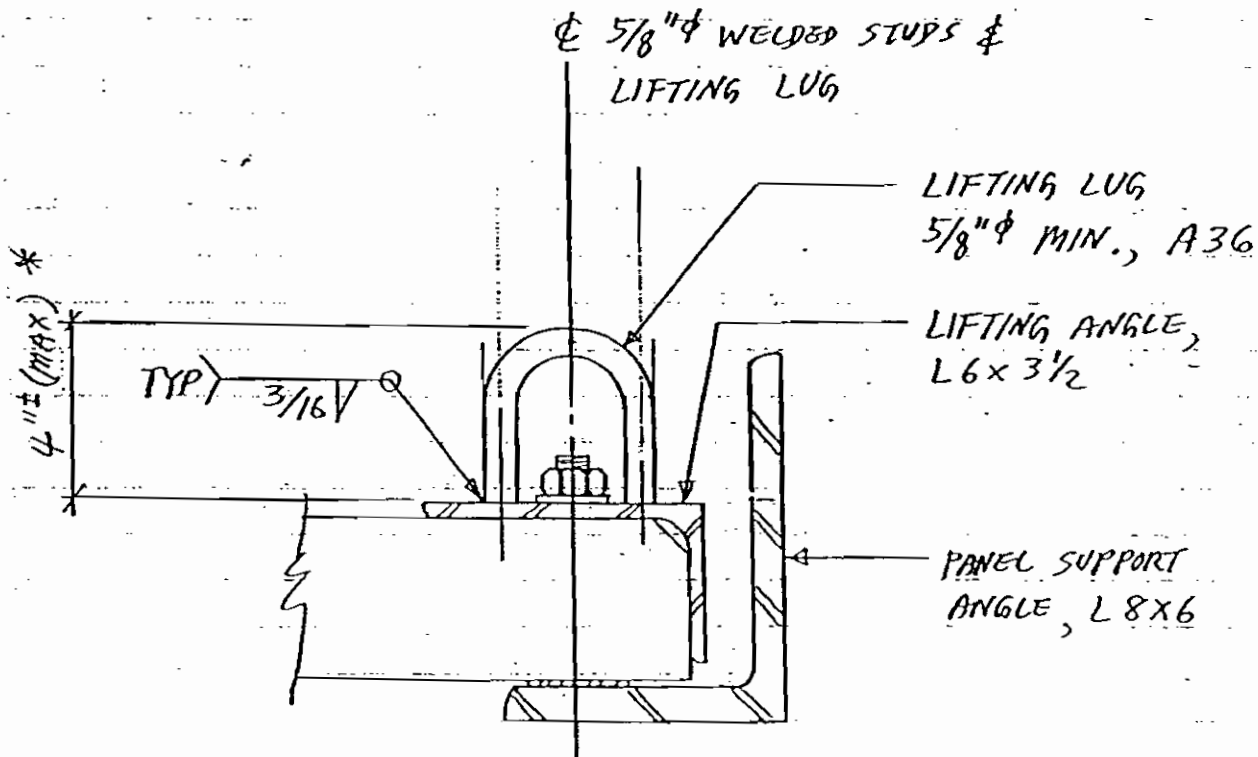
PLAN - PANEL SUPPORT ANGLE

SCARS: 3/16" = 1'-0" HORIZ.

$3/4" = 1'-0"$ VERT

NOTE: DRAWING
ILLUSTRATES 4'x12'
PANELS. 8'x12'
PANELS ARE BEING
CONSIDERED.

ESTIMOTE A JOINT VENTURE	Made By: [REDACTED]	Date: 2/4/97	Job Number
	Checked By:	Date:	QM-081
For CEILING PANEL SUPPORTS	Backchecked By:	Date:	Sheet Number 17



SECTION B-B
SCALE: 3" = 1'-0"

* ADJUST SIZE AS
REQ'D BUT SUBJECT
TO APPROVAL OF
THE ENGINEER

ESTIMATE

A Joint Venture

Made By: [REDACTED]

Date: 2/4/97

Job Number

Checked By:

Date:

QM-081

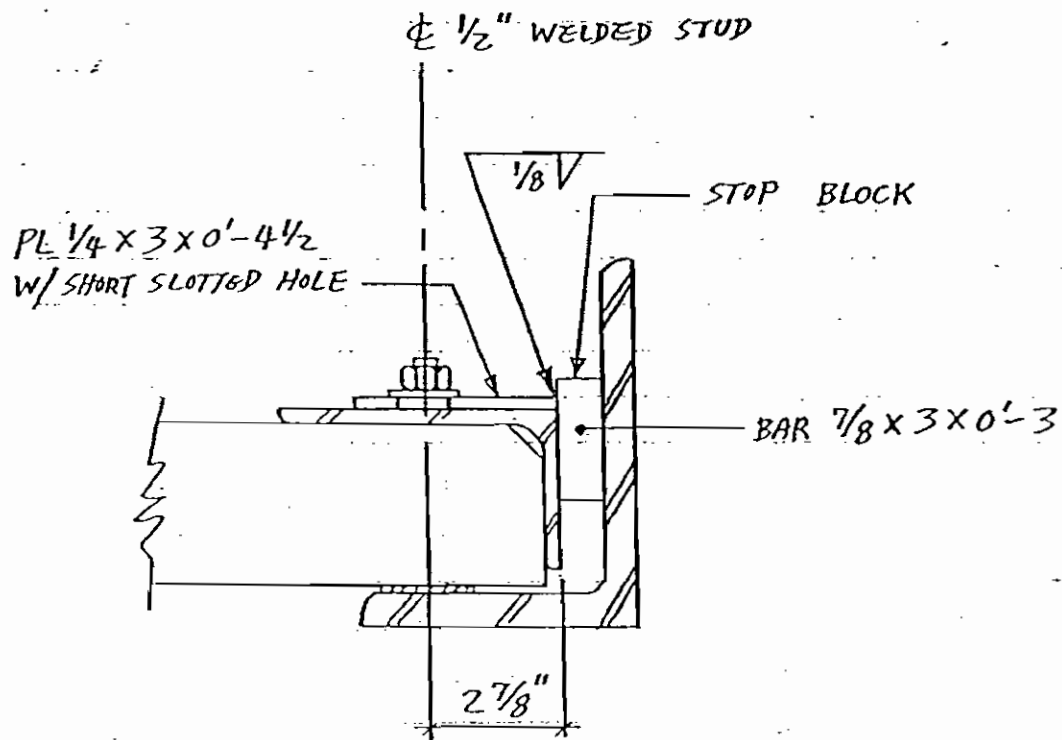
For CEILING PANEL SUPPORTS

Backchecked By:

Date:

Sheet Number

10

SECTION C-C

SCALE: 3" = 1'-0"

Made By: [REDACTED]

Date: 12/20/96

Job Number

Checked By: [REDACTED]

Date: 1/17

QM-081

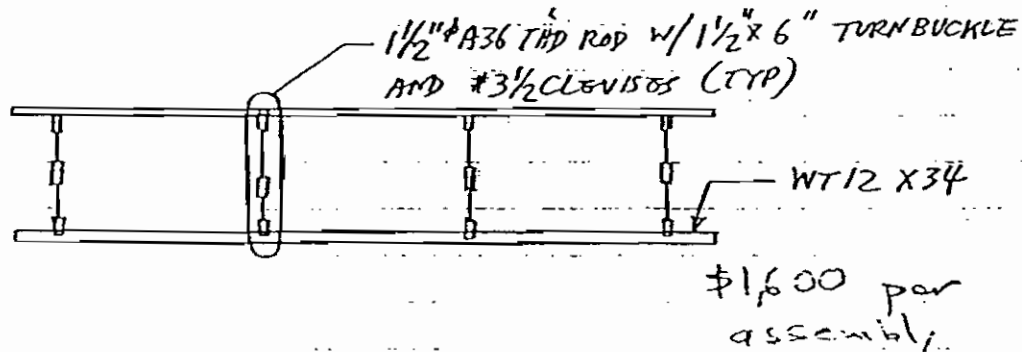
For CEILING SUPPORT SYSTEM

Backchecked By:

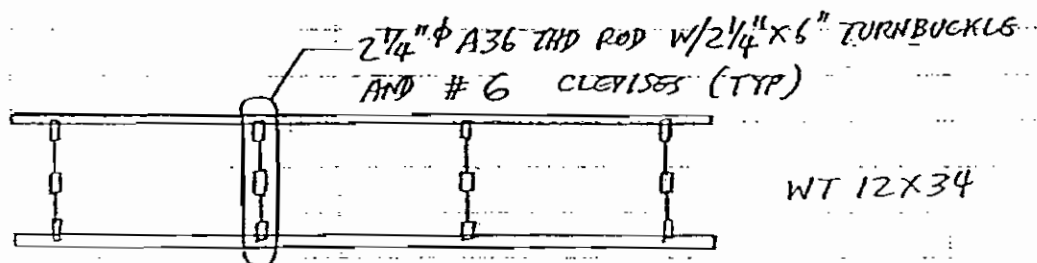
Date:

Sheet Number

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CASE 5-H



CASE 6-H

THE ABOVE RODS, TURNBUCKLES AND CLEVISSES CAN FUNCTION CONTINUOUSLY AT A TEMPERATURE OF 650°F MAX.

ESTIMATE A Joint Venture	Made By: [REDACTED]	Date: 1/16/97	Job Number
	Checked By:	Date:	QM-081
For	Backchecked By:	Date:	Sheet Number 20

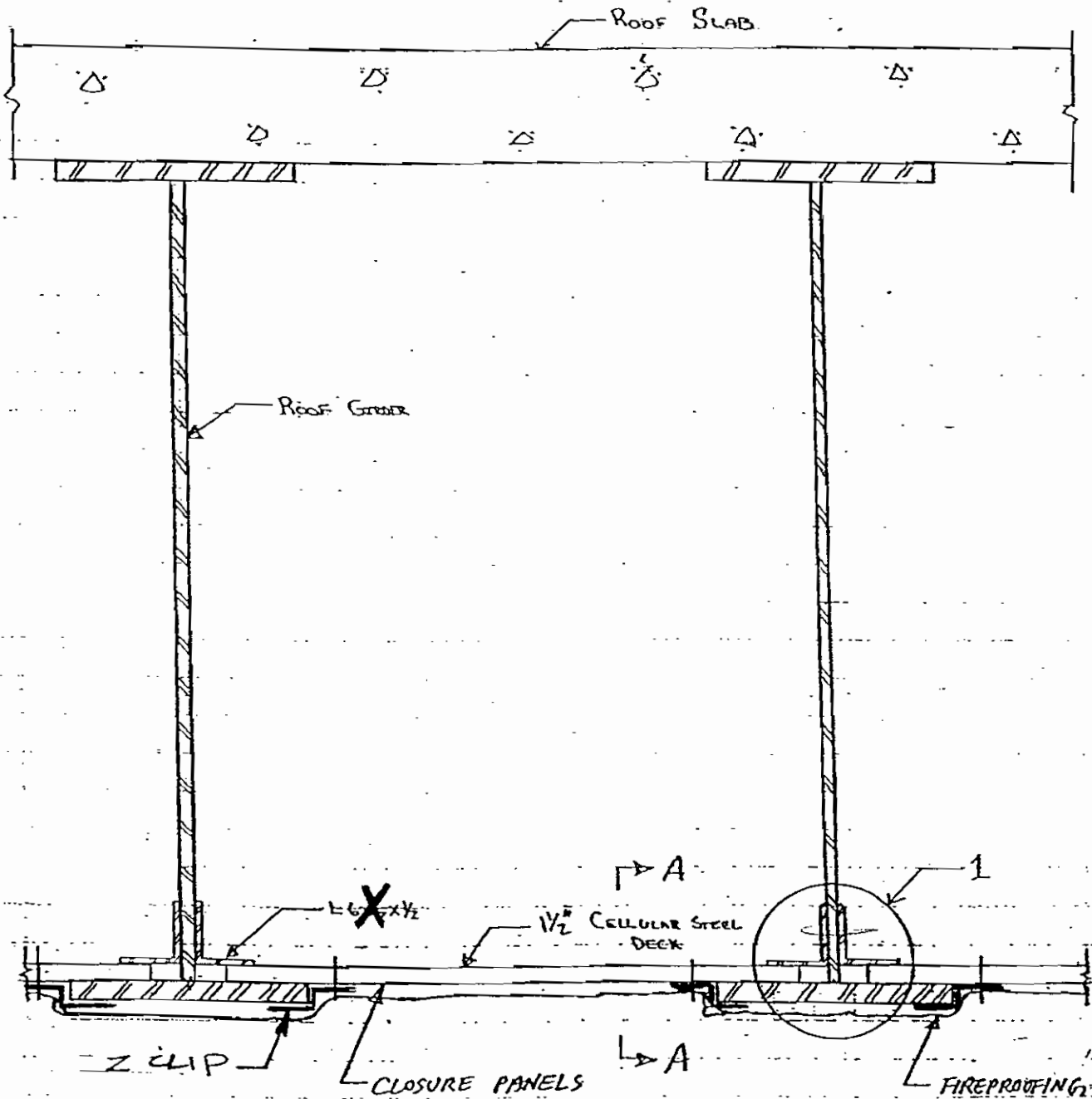
PRELIMINARY DESIGN

SUB-CEILING

CLOSURE PANELS AT BOTTOM FLANGES
OF ROOF GIRDERS

THE FOLLOWING SKETCHES ARE PRELIMINARY DESIGN AND
DETAILS OF THE SUB-CEILING SYSTEM WHICH SHOULD
PROVIDE A 2-HOUR FIRE RATING FOR THE TUNNEL
ROOF AND ITS SUPPORT SYSTEM OF ROOF GIRDERS.

ESTIMATE A Joint Venture	Made By: [REDACTED]	Date: 1/97	Job Number
	Checked By:	Date:	QM-081
	For CLOSURE PANELS @ ROOF GIRDER BOTTOM FLANGES	Backchecked By:	Date:

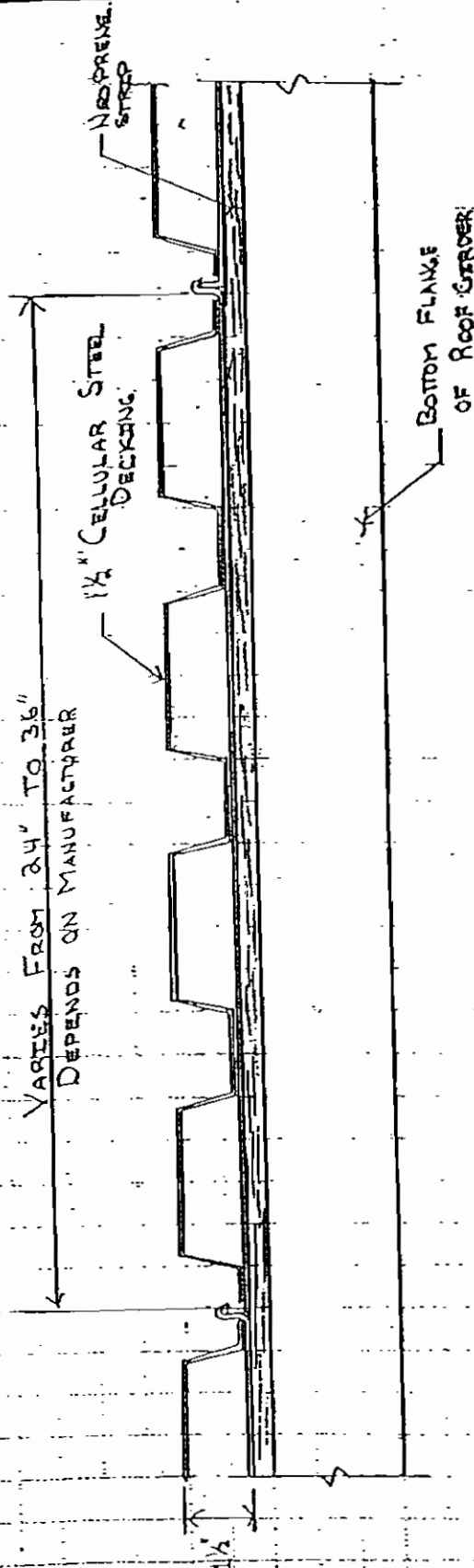


TYPICAL SECTION AT ROOF GIRDERS

SCALE 3/4" = 1'-0"

ESTIMATE A JOINT VENTURE	Made By: [REDACTED]	Date: 1/97	Job Number
	Checked By:	Date:	QM-081
For CLOSURE PANELS @ ROOF GIRDER BOTTOM FLANGES	Backchecked By:	Date:	Sheet Number 22

NOTE: FIRE PROOFING NOT SHOWN
 FIRE PROOFING SHALL BE CONTINUOUS
 AND APPLIED TO BOTTOM OF PANELS.
 THE THICKNESS SHALL PROVIDE
 A 2 HOUR FIRE RATING.



SECTION A
 SCALE 3" = 1'-0"

ESTIMATE

A Joint Venture

Made By: [REDACTED]

Date: 1/97

Job Number

Checked By:

Date:

QM-081

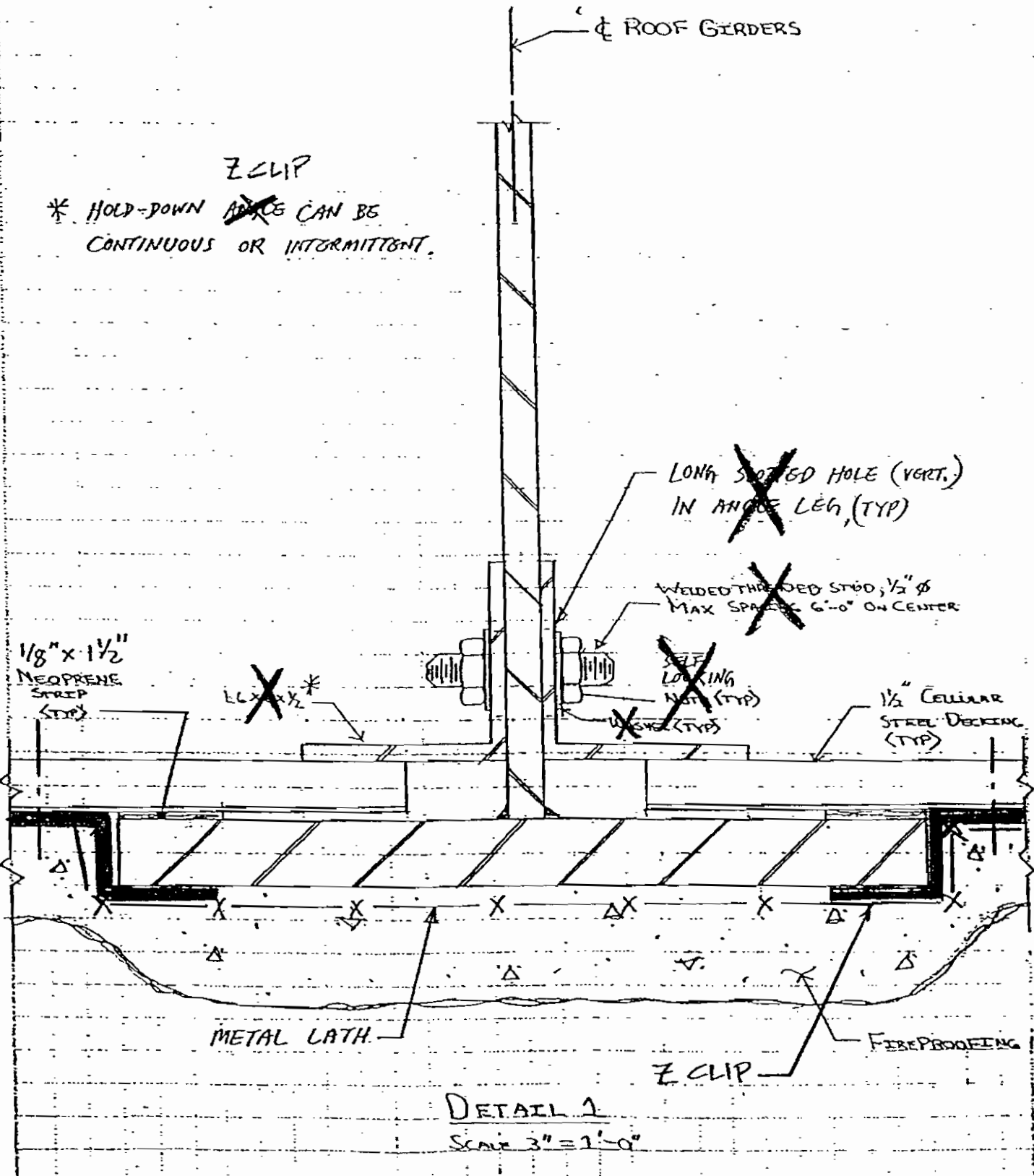
For CLOSURE PANELS @ ROOF GIRDER
BOTTOM FLANGES

Backchecked By:

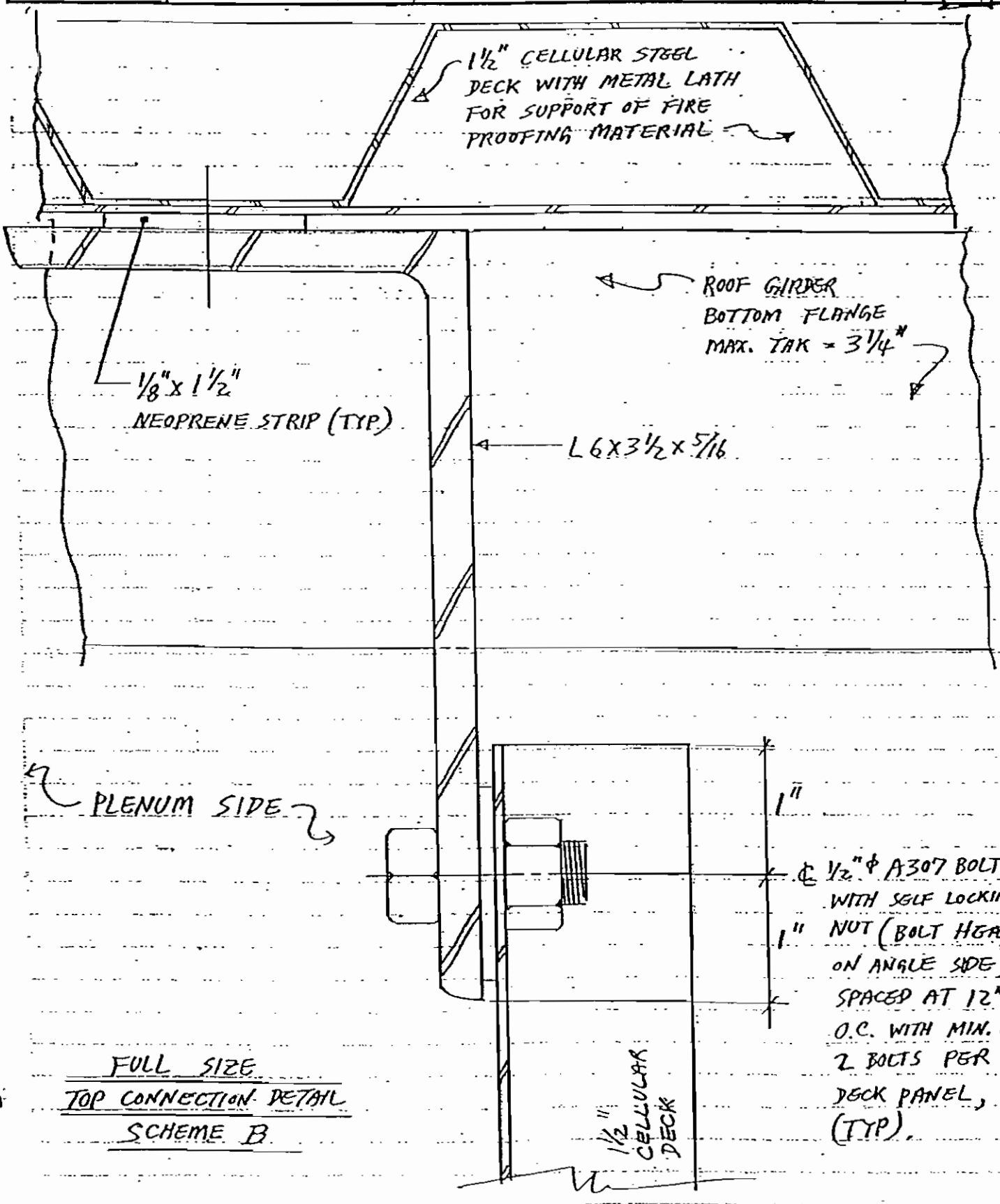
Date:

Sheet Number

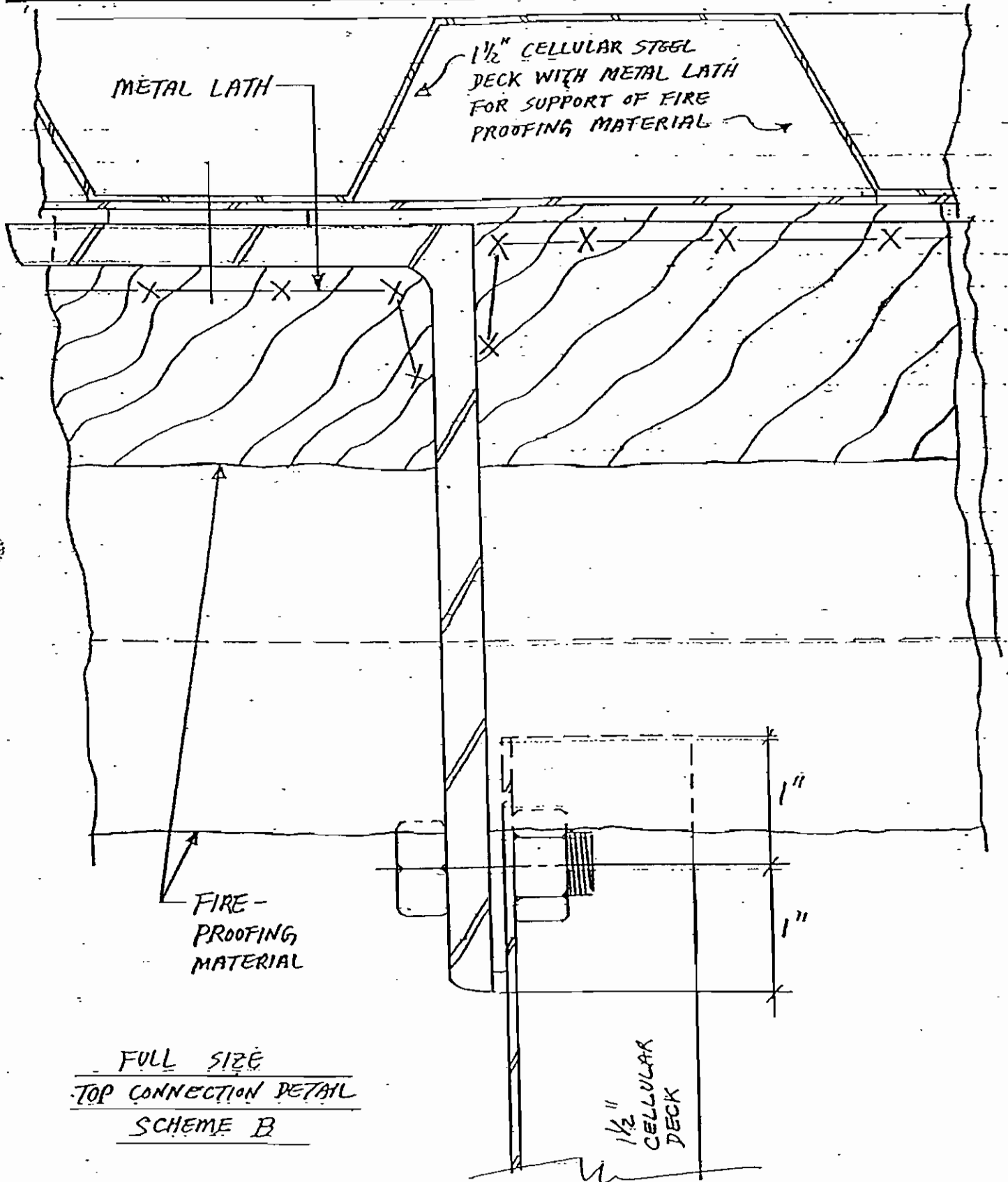
23



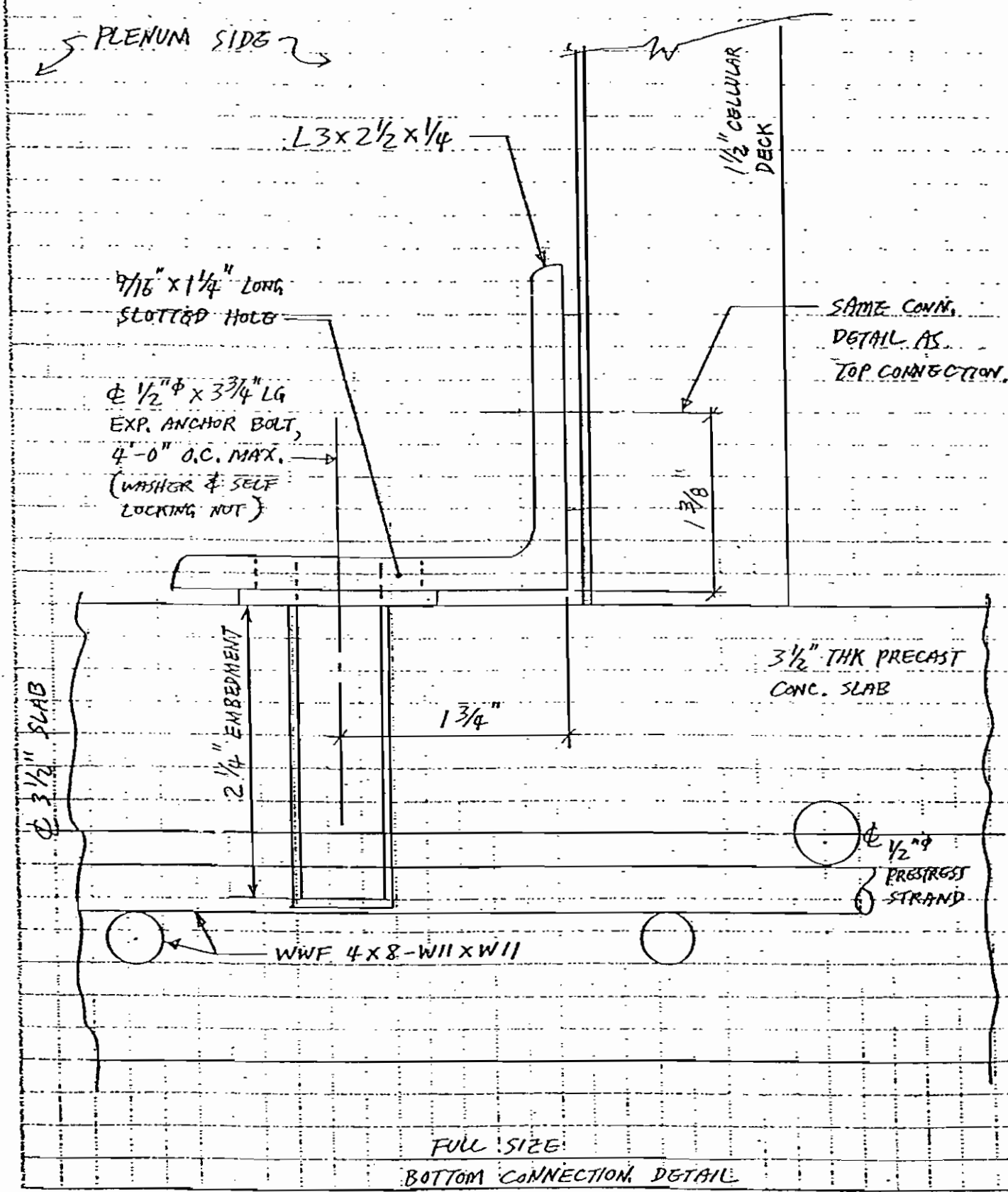
ESTIMATE A JOINT VENTURE	Made By: [REDACTED]	Date: 1/8/97	Job Number
	Checked By:	Date:	QM-081
	Backchecked By:	Date:	Sheet Number 24



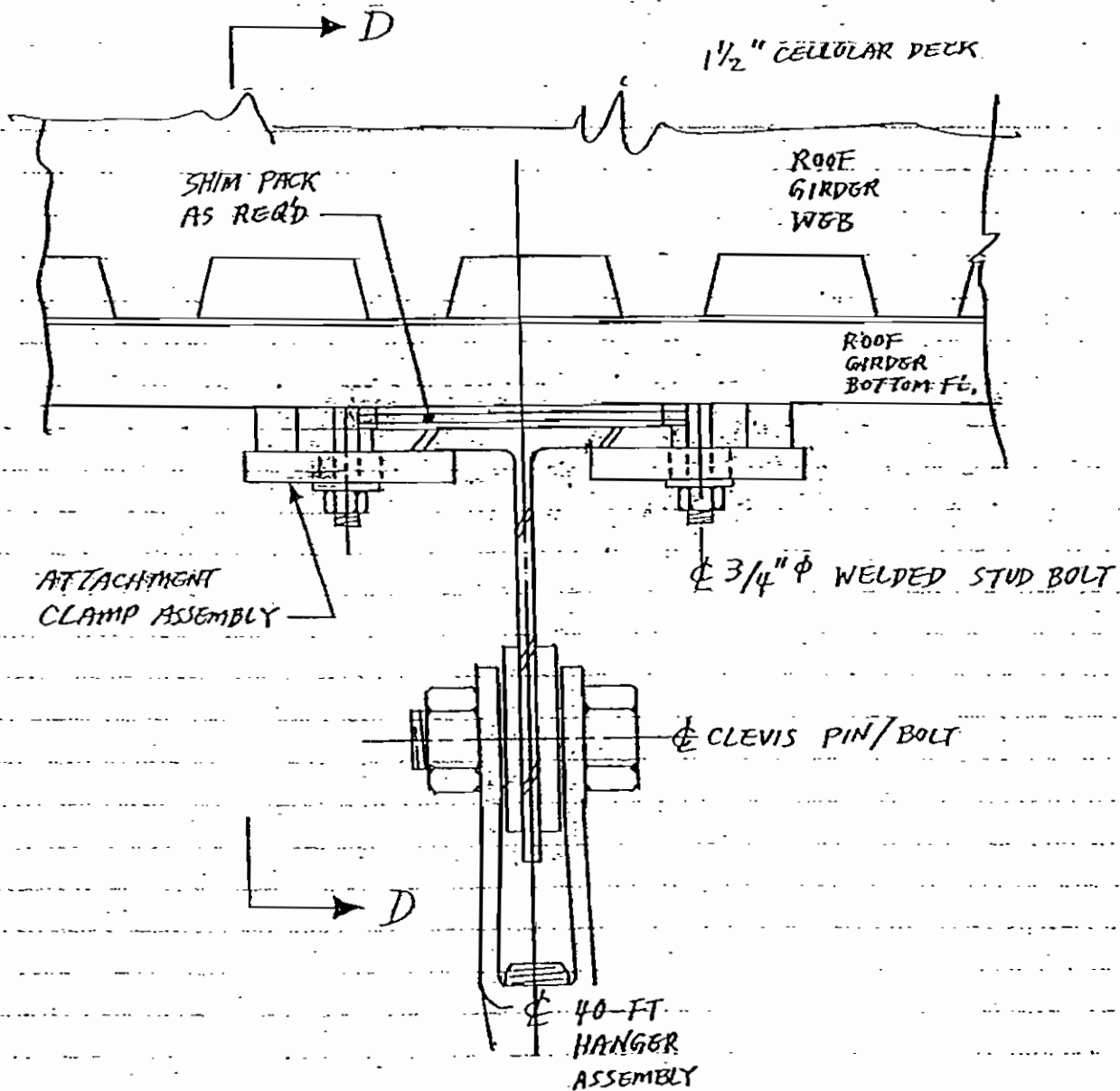
ESTIMOTE A JOINT VENTURE	Made By: XXXXXXXXXX	Date: 1/8/97	Job Number
	Checked By:	Date:	QM-081
	Backchecked By:	Date:	Sheet Number 25



ESTIMATE A JOINT VENTURE For EXHAUST PLENUM-VERT. PANELS	Made By: [REDACTED]	Date: 1/7/97	Job Number
	Checked By:	Date:	QM-081
	Backchecked By:	Date:	Sheet Number 26



ESTIMATE A Joint Venture For CEILING SUPPORT SYSTEM	Made By: [REDACTED]	Date: 1/15/97	Job Number
	Checked By:	Date:	QM-081
	Backchecked By:	Date:	Sheet Number 27



TOP CONNECTOR BEAM OF 40-FT HANGER ASSEMBLY
CONNECTION TO ROOF GIRDER DETAIL
SCALE: 3" = 1'-0"

Made By: [REDACTED]

Date: 1/15/97

Job Number

Checked By:

Date:

Q/M-081

For CEILING SUPPORT SYSTEM

Backchecked By:

Date:

Sheet Number

28

ROOF GIRDER
AND
ATTACHMENT CLAMP ASSEMBLY

NOTE: 1 1/2" CELLULAR
DECK NOT SHOWN.

ROOF GIRDER
BOTTOM FLANGE

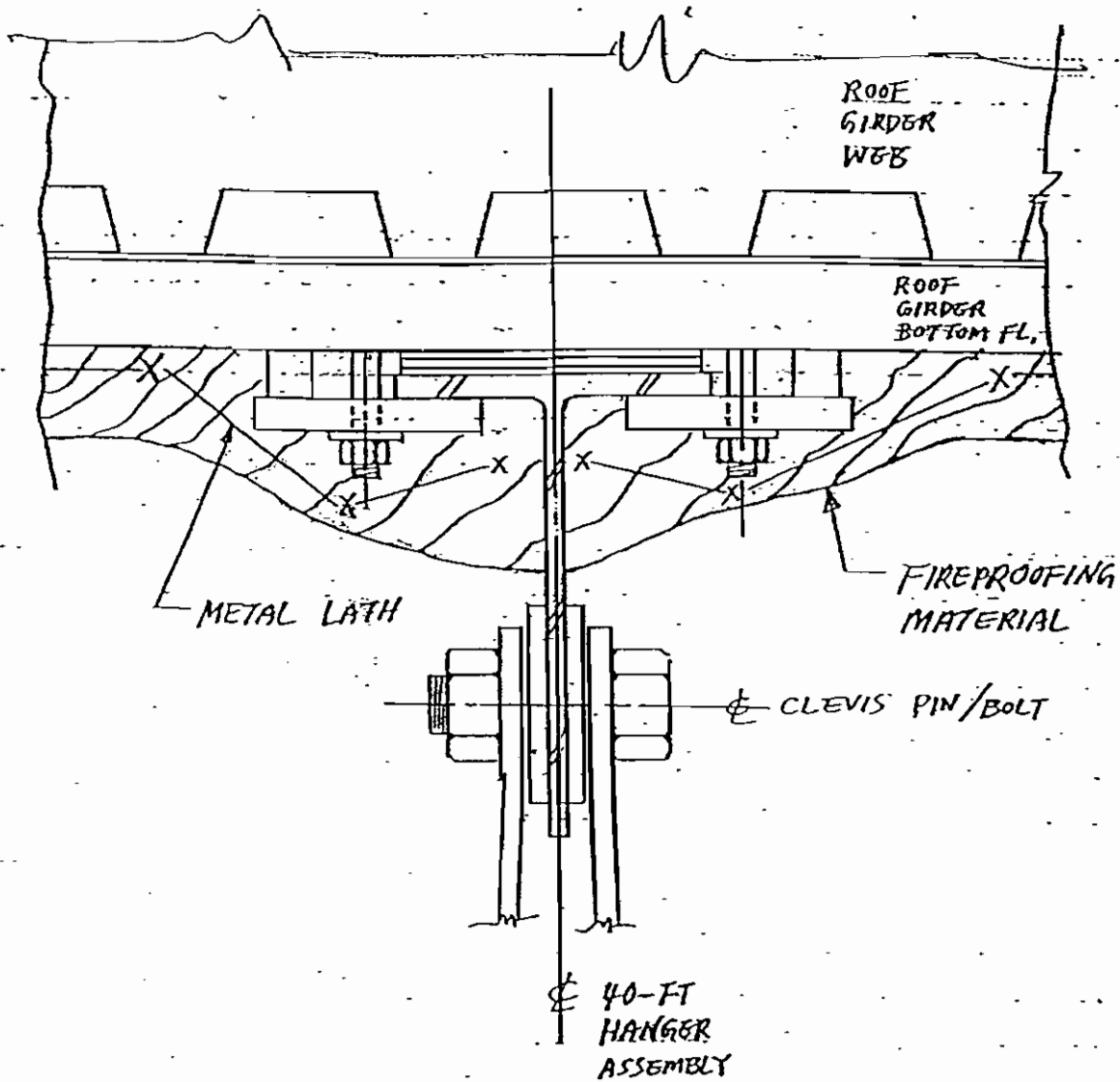
2" 4" 4" 2"

TOP CONNECTOR
OF 40' HANGER
ASSEMBLY

SECTION D

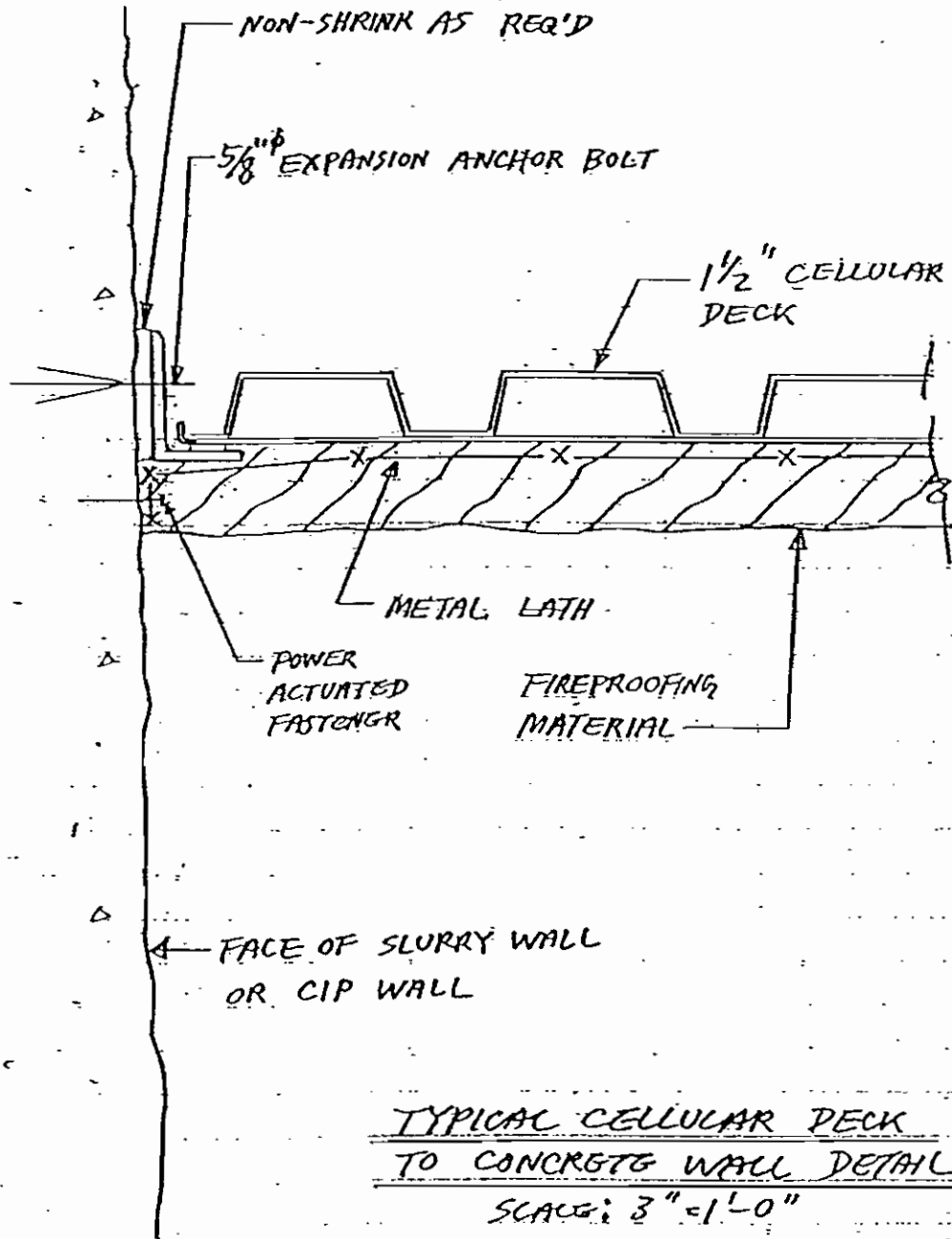
SCALE: 3" = 1'-0"

ESTIMATE A Joint Venture For CEILING SUPPORT SYSTEM	Made By: [REDACTED]	Date: 1/15/97	Job Number
	Checked By:	Date:	QM-081
	Backchecked By:	Date:	Sheet Number 29

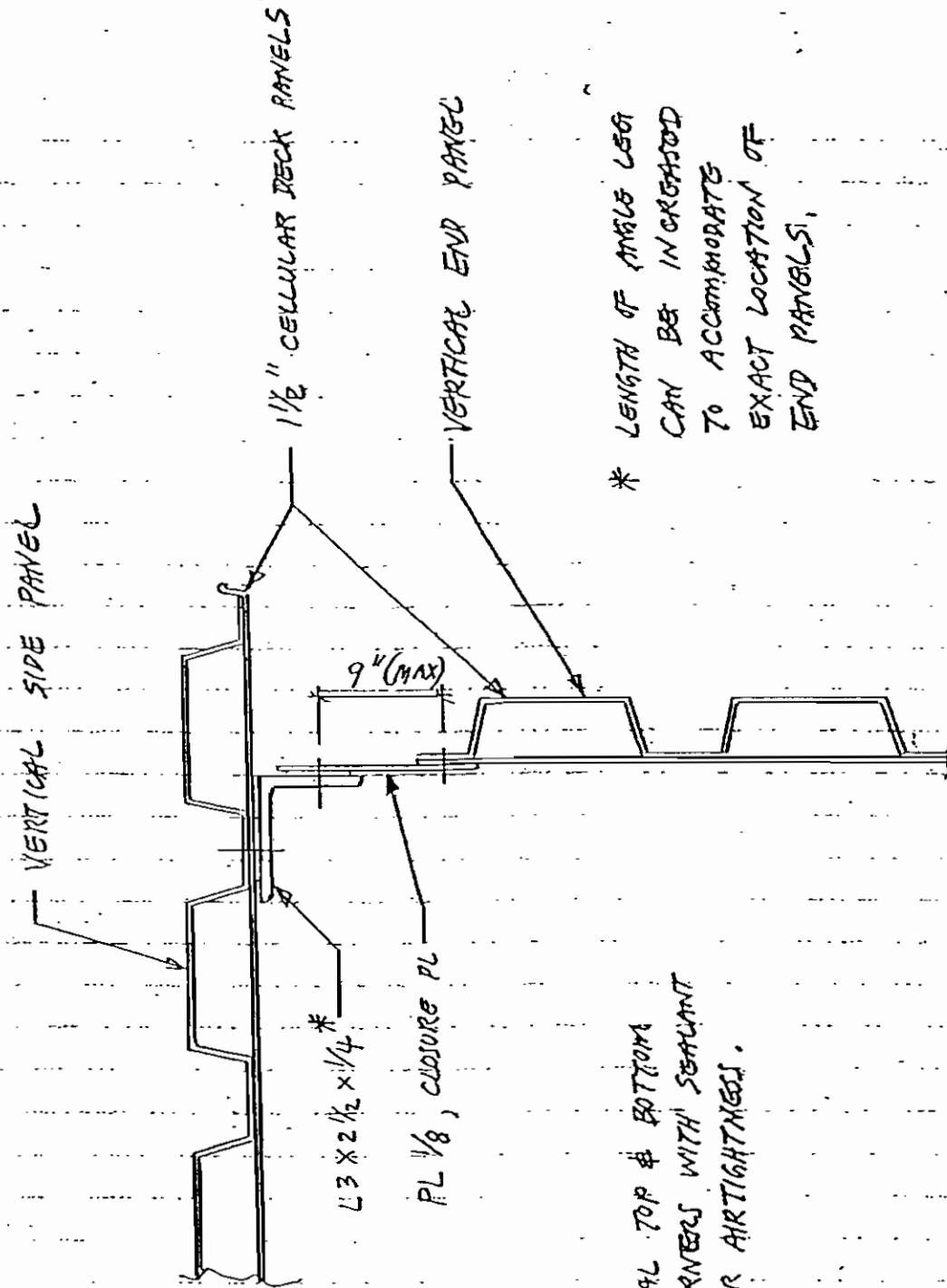


TOP CONNECTOR BEAM - FIREPROOFING DETAIL
 SCALE: 3" = 1'-0"

ESTIMATE A JOINT VENTURE	Made By: [REDACTED]	Date: 1/16/97	Job Number
	Checked By:	Date:	QM-081
For CLOSURE PANELS @ ROOF GIRDER BOTTOM FLANGES	Backchecked By:	Date:	Sheet Number 30



 A Joint Venture	Made By: [REDACTED]	Date: 1/8/97	Job Number
	Checked By:	Date:	QM-081
For EXHAUST PLENUM - VERT. PANELS	Backchecked By:	Date:	Sheet Number 31

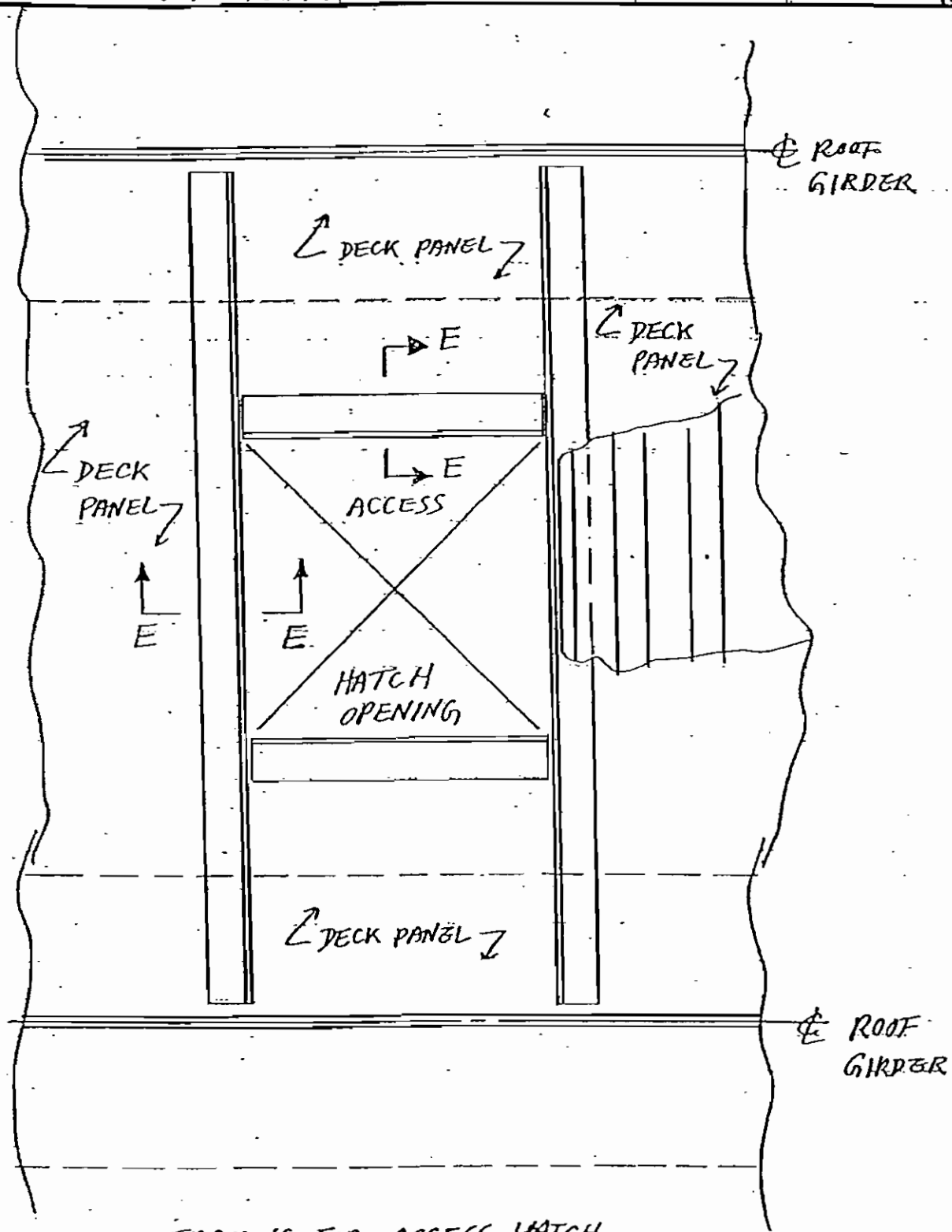


NOTE: SEAL TOP & BOTTOM CORNERS WITH SEALANT FOR AIRTIGHTNESS.

PLENUM
SIDE PANELS
CORNER DETAIL
SCALE: 3" = 1'-0"

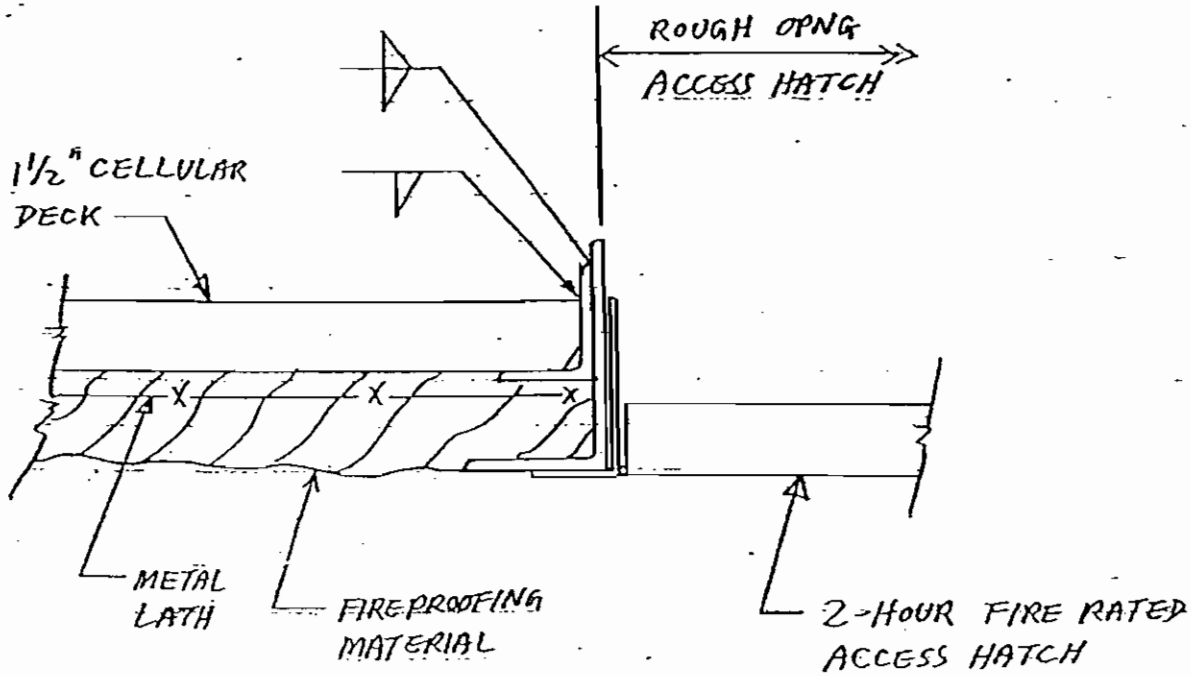
 A JOINT VENTURE	Made By: [REDACTED]	Date: 1/16/97	Job Number
	Checked By:	Date:	QM-081
	Backchecked By:	Date:	Sheet Number 32

For CLOSURE PANELS @ ROOF GIRDER
BOTTOM FLANGES



FRAMING FOR ACCESS HATCH
SCALE: 1" = 1'-0"

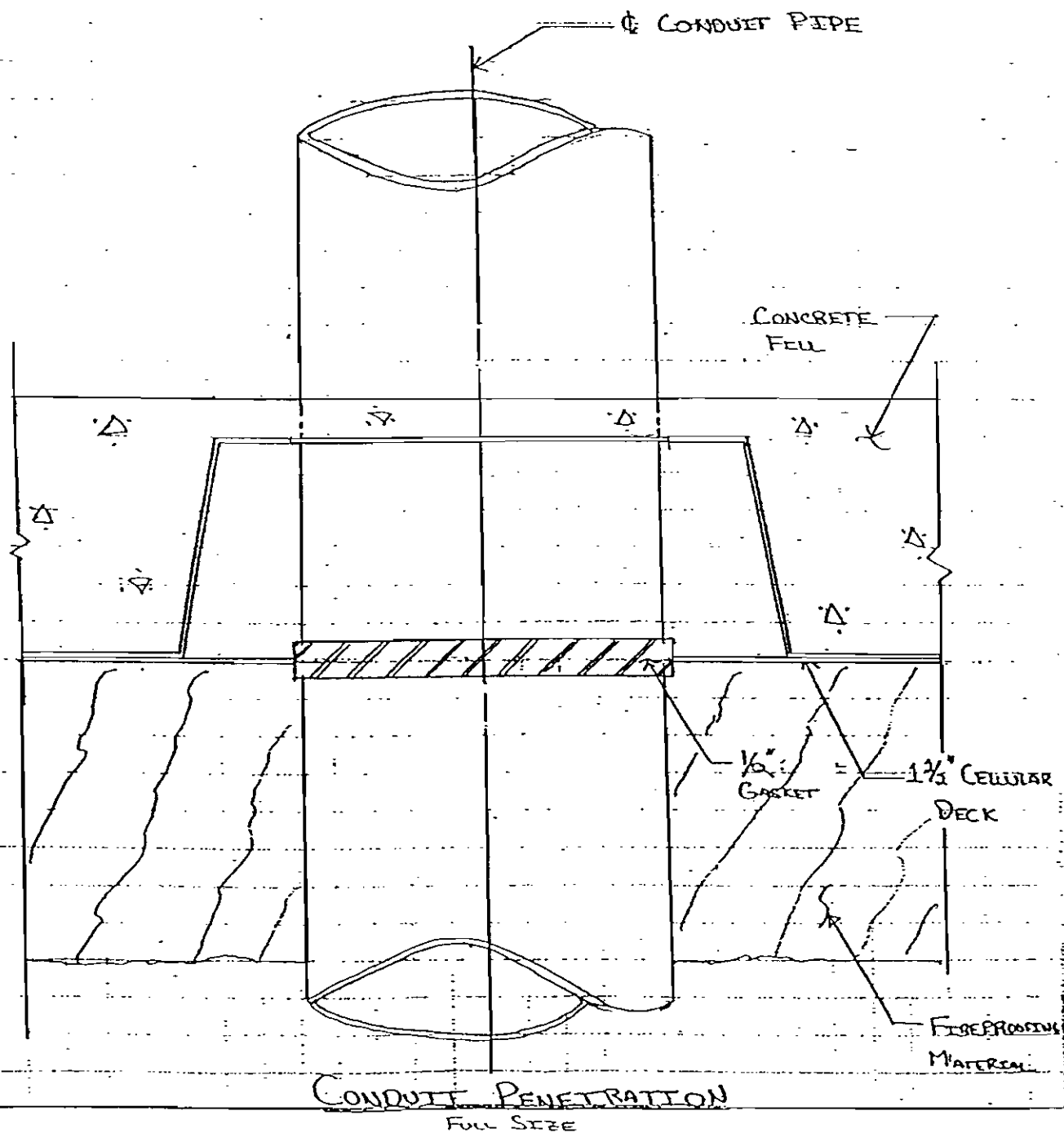
ESTIMATE A Joint Venture	Made By: [REDACTED]	Date: 1/16/97	Job Number
	Checked By:	Date:	QM-081
For CLOSURE PANELS @ ROOF GIRDER BOTTOM FLANGES	Backchecked By:	Date:	Sheet Number 33



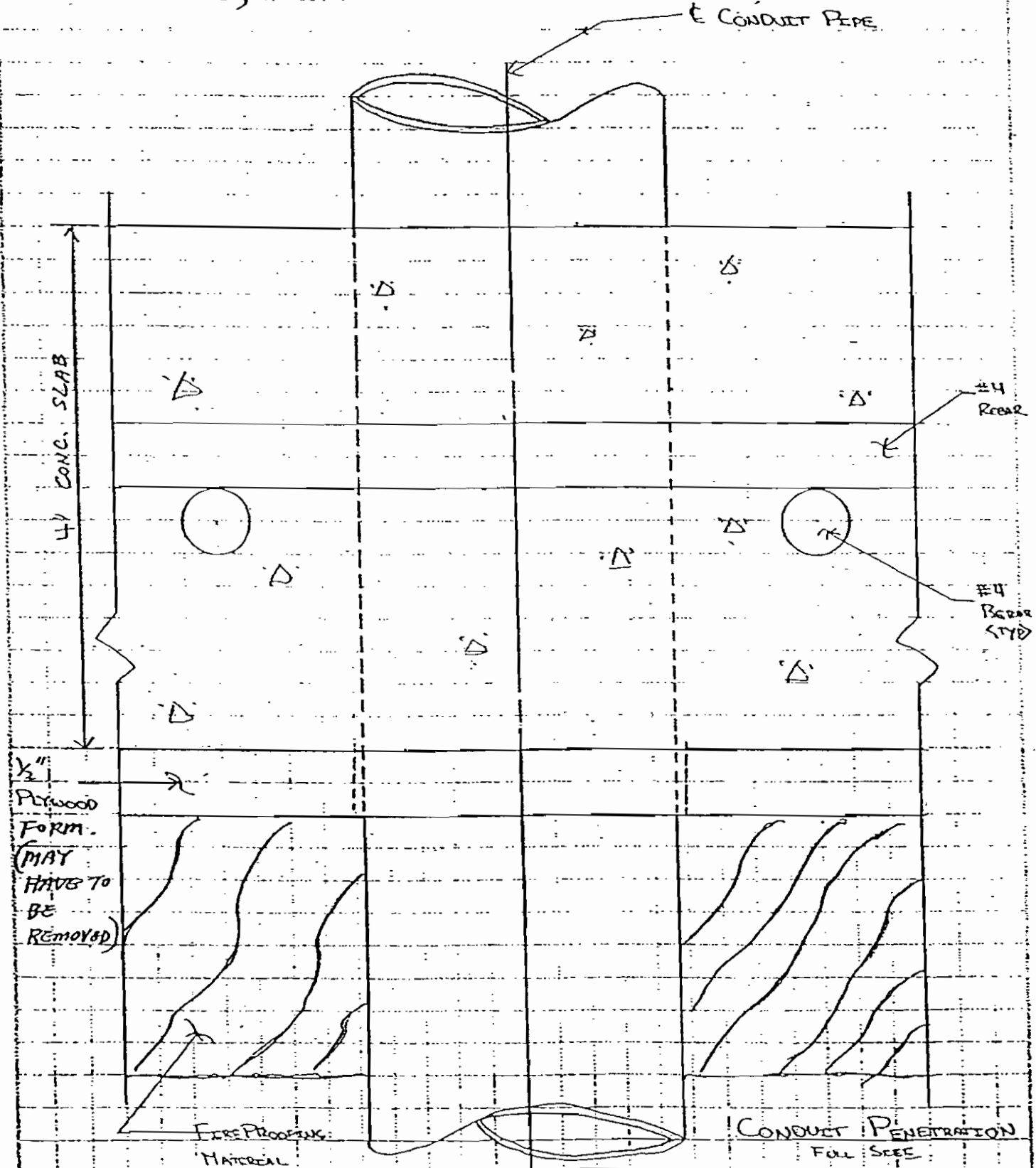
SECTION E
 SCALE: 3" = 1'-0"

ESTIMATE A Joint Venture	Made By: [REDACTED]	Date: 1/97	Job Number
	Checked By:	Date:	QM-081
For CLOSURE PANEL @ ROOF GIRDER'S BOTTOM FLANGE	Backchecked By:	Date:	Sheet Number 34

FULL SIZE DETAIL OF CONDUIT PENETRATION THROUGH THE CLOSURE PANELS, SCHEME A:



FULL SIZE DETAIL OF CONDUIT PENETRATION THROUGH THE
CLOSURE PANELS, SCHEME B:



**SUBJECT: Design Policy Memorandum
No. 107**

REF: J.A. Vears Memo to DPM Manual Holders, dated 3/20/97

Each DPM is intended to be implemented as stated in "C. Contracts Affected".

Therefore, Section Design Consultants receiving these documents are required to return a signed copy of this page to the responsible Authorized Representative, with appropriate responses filled in below. Make copies of this form if necessary to cover more than one construction contract.

SDC RESPONSE

TO:

B/PB Authorized Representative for Contract D _____
Construction Contract C _____ Stage: _____

1. ☐ Received and inserted in the manual.
2. ☐ Will implement within existing Scope of Services.
3. ☐ Not applicable to this Contract.
4. ☐ Applicable but may have cost and schedule effects.
Recommendations attached by letter.

Remarks:

Signed: _____
SDC's Project Manager

Date: _____